A village health worker is a person who helps guide family members and neighbors toward better health. This person is often chosen by the community because he or she is kind, capable, and trustworthy. Some village health workers receive formal training and support from government programs such as the Ministry of Health. Others do not have any official title but are respected members of the community who help people with health problems. Many of them learn by observing others, assisting experienced workers, and studying on their own.

In a broader sense, a village health worker is anyone who contributes to making the village a healthier place. Parents can teach their children the importance of cleanliness. Farmers can cooperate to grow enough nutritious food. Teachers can educate students about preventing and treating common illnesses and injuries. Children can share health knowledge with their families. Shopkeepers can learn about the correct use of medicines and guide customers responsibly. Midwives can advise families about good nutrition during pregnancy, breastfeeding, and family planning. In this way, almost everyone in the village can play a role in improving health.

This book is written for all people who want to improve the health and well-being of themselves, their families, and their communities. It is not only for trained health workers, nurses, or doctors, but for everyone. The knowledge in it should be shared openly. A health worker should use such knowledge to educate others, perhaps by gathering small groups to read and discuss health topics together. A village health worker lives among the people and works at their level. The first responsibility of a health worker is to share knowledge.

To truly improve a village's health, a health worker must understand not only medical needs but also human needs. Compassion and concern for people are just as important as knowledge of medicine and sanitation. Being kind is essential. A friendly word, a smile, or a caring gesture can sometimes help more than medicine. People should be treated as equals, with dignity and respect. Even when busy or stressed, a health worker should remember the feelings and needs of others. It is helpful to imagine how one would treat a member of one's own family in the same situation. Sick people should always be treated as human beings, especially those who are seriously ill or dying, and their families should also receive kindness and support.

A health worker's first duty is to teach. Teaching means helping people learn how to prevent illness and how to recognize and manage common health problems. It includes guiding them in the safe and sensible use of home remedies and common medicines. Health knowledge should not be kept secret. When explained carefully, most health information can help people take better care of themselves. Some professionals may suggest that self-care is risky, but in reality, many common health problems can be managed early and effectively at home if people are properly informed. When people understand how to care for themselves, they become less dependent and more confident.

It is also important to respect the traditions and beliefs of the community. Learning modern medicine does not mean rejecting traditional healing practices. Many traditional methods include valuable human connection and care that should not be lost. However, if a certain custom is harmful, it should be addressed carefully and respectfully. People should never be told simply that they are wrong. Instead, they should be helped to understand why a different approach may be safer or healthier. Change takes time because people hold strong beliefs about what they think is right.

Modern medicine does not have all the answers. While it has solved many problems, it has also created new ones, such as overdependence on medicines and experts. People may forget how to care for themselves and each other. Therefore, a health worker should move slowly and respectfully when encouraging change. The goal is to build on the community's existing knowledge and culture, not to replace it. Traditional healers and midwives should be partners, not opponents. Health workers should learn from them and encourage mutual learning and cooperation.

A health worker must know his or her own limits. No matter how much or how little knowledge and skill a person has, good work can be done if one stays within those limits. This means doing only what one has learned and has enough experience to do safely. A health worker should not attempt procedures that may harm someone if they are not properly trained.

At the same time, judgment is important. Sometimes the decision to act depends on how far away expert help is. For example, if a mother has just given birth and is bleeding more than normal, and a medical center is nearby, it may be best to take her there immediately. However, if she is bleeding heavily and the health center is far away, the health worker may decide to take emergency action, even if not fully trained in that specific procedure. One should not take unnecessary risks, but when doing nothing is more dangerous, it may be necessary to act with care and courage. The priority should always be the safety of the sick person.

A health worker must continue learning. No one should prevent a person from gaining knowledge that can help others. Every opportunity to study books, attend training sessions, or ask experts questions should be used. Learning from doctors, sanitation officers, agricultural specialists, and others can improve the ability to serve the community. Without continuous learning, a health worker will eventually have nothing new to teach.

It is also important to practice what is taught. People pay more attention to actions than words. A health worker should maintain good personal habits and set an example for the community. If encouraging others to build latrines, the health worker's own family should already have one. When organizing community projects, such as digging a garbage pit, the health worker should work just as hard as everyone else.

A health worker should work with joy and enthusiasm. If community improvement is not enjoyable, others will not feel motivated to participate. Community projects can be made enjoyable by turning them into shared events with music, food, and celebration. Even hard work becomes easier when done together in a positive spirit. Whether paid or unpaid, a health worker should never refuse care to someone who is poor. Respect and love from the community are more valuable than money. The main purpose should be to serve people, not to earn income.

A responsible health worker looks ahead and works to prevent illness before it begins. Many diseases can be prevented if their causes are understood and addressed early. Health problems usually have several connected causes, and lasting solutions require identifying and correcting the root causes.

For example, diarrhea may be a leading cause of death among young children in some villages. Poor sanitation and hygiene contribute to its spread, and building latrines and teaching cleanliness can help reduce it. However, children who are poorly nourished are more likely to die from diarrhea because their bodies are too weak to fight infection. Therefore, preventing malnutrition is also essential.

Poor nutrition itself may have many causes. It may result from lack of knowledge about proper feeding, lack of land or income, unfair distribution of resources, poor farming practices, large family size, misuse of money, or lack of planning for the future. Many social and economic factors can influence health. A health worker should help the community understand these deeper causes and encourage collective solutions.

Preventing deaths from diarrhea requires more than clean water and oral rehydration solutions. Long-term solutions may include better child spacing, improved land use, fairer sharing of resources, and stronger cooperation within the community. Greed and short-sighted thinking often contribute to suffering. Health workers should encourage people to share, cooperate, and plan for the future.

Health is influenced by many factors such as food production, education, land distribution, and social relationships. Good health is more than the absence of disease. It includes physical, mental, and social well-being. A healthy community is one where people trust each other, work together, share resources, support each other during difficulties, and help each other grow. While solving daily problems is important, the greater responsibility of a health worker is to help create a healthier and more caring community.

A health worker should take a careful look at the community. Growing up in the same place gives an inside understanding of many local health problems. However, to understand the full situation, it is necessary to observe the community from different perspectives. A health worker's responsibility is to care for the well-being of all people, not just close friends or those who come seeking help. This requires visiting homes, fields, schools, and common gathering places. By spending time with people, the health worker can understand their happiness, struggles, habits, and daily practices that either improve health or cause sickness and injury.

Before starting any community project, it is important to think carefully about what it will require and whether it is likely to succeed. Several factors must be considered. First are the felt needs, which are the problems people believe are most important. Second are the real needs, which are the actual steps required to solve problems in a lasting way. Third is the willingness of people to take action and make changes. Fourth are the available resources, including people, skills, materials, and money.

For example, if a heavy smoker complains of a worsening cough, his felt need is relief from the cough. However, his real need is to stop smoking. To solve the problem, he must be willing to give up smoking and understand why it is harmful. Resources that may help him include clear information about the dangers of smoking and support from family, friends, and the health worker.

To identify community needs, a health worker should gather information about major health concerns and general problems affecting people's lives. Preparing a short and simple list of questions can help guide discussions. These questions should encourage people not only to provide information but also to reflect on their own situation. The list should not be too long or complicated, especially when visiting homes. People should never feel like numbers in a survey. The main focus should always be on understanding individual feelings and concerns. Sometimes it may be better not to carry a written list but to keep important questions in mind during conversations.

When assessing health needs, it is useful to consider daily living conditions, housing, sanitation, education, and population factors. Housing conditions such as building materials, cleanliness, ventilation, sleeping arrangements, and cooking practices can affect health. The presence of pests like flies, rats, and insects may spread disease. Food protection and animal contact within homes may also influence health risks. Water sources should be examined to determine whether they are safe for drinking. The availability and proper use of latrines and garbage disposal practices are important for sanitation.

Population factors also affect health. The number of people in the community, the number of children, literacy rates, and school quality all influence well-being. Information about births, deaths, causes of illness, and common injuries helps identify major health threats. Chronic illnesses and patterns of sickness over the year provide insight into long-term problems. Family size, child deaths, and underlying causes such as malnutrition or infection should also be considered. Interest in family planning and reasons behind it may reveal social and economic concerns. By understanding these interconnected factors, a health worker can help the community think critically about its health and plan meaningful improvements.

Nutrition is an important part of community health. It is useful to know how many mothers breastfeed their babies and for how long. Breastfed babies are often healthier because breast milk provides complete nutrition and protection against infection. It is also important to understand the main foods people eat, where those foods come from, and whether families make full use of available resources. Identifying children who are underweight or show signs of malnutrition helps determine the seriousness of nutritional problems. Knowledge about nutrition among parents and schoolchildren also affects health outcomes. Habits such as smoking and frequent consumption of alcohol or sugary drinks can harm both individuals and their families.

Land and food production are closely linked to nutrition. A community must consider whether the land produces enough food for every family and whether it will continue to do so as the population grows. The distribution of farmland and ownership patterns may influence food security. Efforts to improve agricultural productivity, as well as safe storage of crops to prevent loss or damage, are also important for maintaining an adequate food supply.

Healing practices and available health services should be evaluated. Local midwives and traditional healers often play a significant role in care. Some traditional practices may be valuable, while others may be harmful and need careful improvement. Nearby health services, their quality, cost, and accessibility influence how people seek treatment. Vaccination coverage and other preventive measures also indicate how well a community protects itself from disease.

Self-help is another key aspect of health. A community should consider which factors most affect its well-being now and in the future. Many common health problems can be managed at home if people have proper knowledge. However, some situations require outside assistance. It is important to understand whether people are interested in improving self-care practices and what barriers may exist. Social fairness and equality also affect health. The rights and treatment of rich and poor people, men and women, and children influence overall well-being. Cooperation, sharing, and mutual support strengthen a community's ability to face health challenges.

Using local resources is often the most sustainable way to meet health needs. Some programs require outside supplies, such as vaccines. However, many improvements can be made using local materials and skills. Fencing water sources, building simple latrines, or encouraging breastfeeding are examples of actions that rely mainly on community effort. Local resources are often more dependable, affordable, and appropriate. Breast milk is a prime example of a high-quality local resource that protects infant health without cost.

After examining needs and resources, the community must decide which actions are most important. Poor nutrition frequently contributes to many other health problems. If hunger and malnutrition are widespread, improving nutrition should be a priority. There are various approaches to improving food security, such as family gardens, better farming methods, crop rotation, irrigation, fish farming, beekeeping, composting, improved storage techniques, and family planning to manage population growth. Some solutions provide quick benefits, while others require long-term effort.

When trying new ideas, it is best to begin with small experiments. Starting small reduces risk and allows learning from both success and failure. For example, if a new type of bean is introduced, a small area can be planted under different conditions to see whether it grows well and whether people are willing to eat it. When experimenting, only one condition should be changed at a time, such as soil type, fertilizer amount, water level, or seed variety. This method helps identify which factor truly affects the outcome. Learning through careful experimentation enables communities to make informed decisions and improve their health and nutrition gradually.

Health depends largely on whether people have enough food. Since most food comes from the land, the way land is used directly affects health. When land is managed well, it can produce more food, but every piece of land has limits. It can only support a certain number of people. In many places, farming families do not have enough land to meet their needs. As populations grow and families have many children, the pressure on limited land increases, and hunger becomes more common.

Many health programs try to create a balance between people and land through family planning. The idea is that smaller families will mean more food and resources for each person. However, family planning alone does not solve the problem. Poor families often prefer to have many children because children help with work and may support parents in old age. In areas where child death is common and social security systems do not exist, having many children is seen as a form of security.

Some groups believe that hunger is not caused by too little land but by unfair distribution of land and wealth. In many places, most land is owned by a few people. When land and resources are shared more fairly and families feel economically secure, they usually choose to have fewer children. True balance between people and land is more likely when there is social justice and economic security, rather than relying only on family planning programs.

Health work also requires a balance between treatment and prevention. People often first seek relief for sickness and suffering. Therefore, a health worker must respond to immediate needs through treatment. At the same time, it is important to help people understand how diseases can be prevented. If a health worker focuses only on prevention and ignores present suffering, people may not cooperate. Treatment and prevention should work together. Early treatment can prevent mild illnesses from becoming severe.

A health worker should begin where the people are and respect their current beliefs and attitudes. While treating illness, the worker can gently introduce ideas about prevention. For example, when treating a child with worms, the worker should explain both how to treat the infection and how to prevent it in the future. Home visits can support families in improving their self-care practices.

An important part of prevention is teaching the sensible and limited use of medicines. Some modern medicines are life-saving, but most minor illnesses do not require medication. The body can often recover with rest, good nutrition, and adequate fluids. People may request medicines even when they are unnecessary. Giving medicine just to satisfy expectations can create dependence and misunderstanding. It is better to explain why medicine is not needed and what simple steps can help recovery.

Common conditions such as colds, mild coughs, and most cases of childhood diarrhea usually do not require antibiotics or strong medicines. Colds improve with rest and fluids. Minor coughs often improve with increased water intake and warm steam. In most cases of childhood diarrhea, the most important treatment is giving plenty of fluids and continuing to feed the child. Many commonly used medicines for diarrhea are unnecessary or even harmful. The key to recovery is proper care, especially by the mother.

Overuse of medicines is harmful for several reasons. It wastes money that could be spent on food or other necessities. It creates dependence on drugs that may not be needed. All medicines carry some risk of side effects. Frequent misuse of antibiotics can cause bacteria to become resistant, making serious infections harder to treat. When important antibiotics are overused for minor illnesses, they may lose effectiveness against dangerous diseases.

Limiting the use of medicines requires educating people so they understand when medicines are necessary and when they are not. Strict rules alone are not enough. When communities are informed and aware, they are more likely to use medicines carefully and responsibly.

Health depends on many things, but most of all it depends on having enough food. Most food comes from the land. When land is used properly, it can produce more food. Health workers should understand how to help land produce enough food for people now and in the future. However, even the best land can only support a limited number of people. Many farming families today do not have enough land to meet their needs or stay healthy.

In many parts of the world, this situation is becoming worse. Families often have many children, so each year there are more people to feed, but the land available to poor families does not increase. Some health programs try to solve this problem through family planning, encouraging families to have fewer children. The idea is that smaller families will mean more food and land for everyone. However, family planning alone does not solve the problem. Poor families often want many children because children help with work and may earn money as they grow older. When parents become old, their children may care for them. For poor families, having many children often feels necessary for survival, especially in places where many children die young. For many people, children are their only form of social security.

Some groups believe that hunger exists not because there is too little land, but because land and wealth are unfairly distributed. A small number of people own most of the land. These groups work to create a fairer distribution of land and wealth so that people can gain more control over their health and lives. When land and wealth are shared more fairly and families feel economically secure, they usually choose to have fewer children. Family planning works best when it is a free and informed choice. A true balance between people and land is more likely when there is social justice and fair distribution of resources.

Health workers should support efforts toward fairness and justice, helping communities build a better balance between population and resources.

There must also be a balance between treatment and prevention. Health workers must respond to people's immediate needs while also thinking about long-term health. People usually first want relief from sickness and pain. Therefore, health workers must help treat illness. At the same time, they should help people understand that many illnesses can be prevented and that they can take action to protect their health.

Sometimes health workers focus too much on prevention and ignore people's current suffering. If they fail to address immediate needs, people may not cooperate with preventive programs. Treatment and prevention should support each other. Early treatment can prevent illness from becoming serious. Teaching families to recognize common health problems and treat them early at home can reduce suffering.

Health workers should begin where people are and work according to their attitudes and beliefs. Over time, as diseases are controlled and understanding grows, communities may naturally focus more on prevention. When a child is sick, a mother needs help with treatment. While helping her, the health worker can also explain how to prevent similar problems in the future. Treatment can be used as an opportunity to teach prevention.

One important part of prevention is teaching people to use medicines wisely and only when needed. Some modern medicines are lifesaving and very important. However, many illnesses get better on their own with rest, good nutrition, and plenty of fluids. Giving medicine when it is not needed can create dependence and misunderstanding. People may believe the medicine cured them when in fact their bodies healed naturally.

Instead of giving unnecessary medicines to please people, health workers should explain why they are not needed and teach families how to care for themselves. This encourages reliance on local resources and protects people from the risks of medicine. Every medicine carries some risk.

Medicines are often overused for common problems such as colds, minor coughs, and diarrhea. Colds usually improve with rest and fluids, and antibiotics do not help. Many coughs improve with drinking plenty of water and breathing warm vapor. Most childhood diarrhea improves with fluids and continued feeding, and many medicines for diarrhea may even be harmful. The most important factor in recovery is proper care at home, especially by the mother.

Overuse of medicines is wasteful and can be dangerous. Money spent unnecessarily on medicines could be used for food. Overuse can cause side effects and may make medicines less effective over time. When powerful medicines are used too often for minor problems, bacteria can become resistant, making serious infections harder to treat. For these reasons, medicines should be used carefully and only when truly needed. The best way to prevent overuse is to educate people so they can make informed decisions.

Home remedies are used everywhere in the world. Many traditional healing methods have been passed down for generations. Some home remedies are very helpful, some are less useful, and some may be harmful. Like modern medicines, they should be used carefully.

For many common illnesses, traditional remedies work as well as modern medicines and are often cheaper and safer. Herbal teas for coughs and colds can be more helpful and safer than strong medicines. For diarrhea in babies, simple fluids such as rice water, teas, or lightly sweetened drinks are often safer and more effective than many modern medicines. The most important treatment is giving enough fluids.

However, some serious diseases require modern medical treatment. Illnesses such as pneumonia, tetanus, typhoid, tuberculosis, appendicitis, infections spread through sexual contact, and fever after childbirth should be treated promptly with modern medicine. Time should not be wasted trying only home remedies in such cases.

Sometimes modern health practices are better, but sometimes traditional methods are more caring and effective. For example, breastfeeding is healthier and safer for babies than artificial feeding, despite past advertising that suggested otherwise. Many babies suffered when mothers were convinced to use bottle feeding instead of breast milk.

Some home remedies work because people believe in them. The power of belief can be strong. When a person trusts a remedy, they may feel better even if the remedy has no direct physical effect. Belief can help reduce fear and stress, which may improve certain conditions such as anxiety, stress-related pain, asthma, indigestion, migraines, or other problems influenced by emotional factors. The care and kindness of the healer also play an important role.

In some cases, belief may even help in physical conditions. For example, many cultures have traditional treatments for snakebite. Although these remedies may not neutralize snake poison, they can calm the person, slow their pulse, and reduce movement. This may slow the spread of poison in the body. However, such remedies are limited and cannot replace proper medical treatment. For poisonous snakebites, modern treatment with antivenom is usually necessary and should be prepared in advance.

Belief can help people heal, but it can also make them sick. When a person strongly believes that something will harm them, fear itself can cause illness. For example, a woman who had a miscarriage was advised to drink orange juice because vitamin C helps strengthen blood vessels. Although the juice was safe, she believed it would harm her. Her fear became so strong that she became very ill, even though there was no physical cause. Later, she was given an injection of distilled water, which has no medical effect. Because she believed injections were powerful, she quickly recovered. The orange juice did not harm her. Her fear made her sick, and her belief in the injection helped her recover.

Many people continue to believe false ideas about witchcraft, injections, and diet. These beliefs can cause unnecessary suffering. Although giving the harmless injection seemed to help, it also strengthened her false belief. Later, the health worker explained the truth to her so she could understand that fear had caused her illness and relief from fear had helped her recover. Understanding reduces fear, and freedom from fear supports health.

Belief in witchcraft, black magic, or the evil eye can also cause illness if a person strongly believes they are bewitched. Fear and anxiety can produce real physical symptoms. A person who thinks they are bewitched may become sick because of their own fear. Witches have no real power except the power people give them through belief. Some strange or serious illnesses, such as tumors or liver disease, may be wrongly blamed on witchcraft, but these diseases have natural causes. Seeking magical cures or revenge will not solve health problems. Serious illness requires proper medical care.

Some common beliefs can be examined carefully. Harming someone believed to be a witch does not help a sick person. When the soft spot on a baby's head sinks inward, it often means the baby is dehydrated and needs fluids quickly to survive. Illnesses or deformities in babies are not caused by eclipses but may result from poor nutrition, lack of iodine, certain medicines, or other medical reasons.

Soft lighting during childbirth may be comfortable, but enough light is needed so the birth attendant can see clearly. A newborn's cord stump should be kept dry until it falls off, but the baby can still be gently cleaned with a damp cloth. Mothers should bathe with warm water soon after childbirth to prevent infection. Avoiding bathing for many weeks can increase the risk of illness.

Breastfeeding is healthier than bottle feeding because breast milk provides better nutrition and protects against infection. After childbirth, women should not avoid nutritious foods. Instead, they should eat a balanced diet including fruits, vegetables, grains, beans, milk, eggs, and meat if available.

Bathing a sick person daily with warm water is helpful and does not cause harm. Fruits and fruit juices are beneficial during colds or fever and do not cause congestion. When a person has a high fever, heavy clothing and covers should be removed to allow air to cool the body. Tea made from willow bark can reduce fever and pain because it contains a natural substance similar to aspirin.

The fontanel is the soft spot on the top of a newborn baby's head where the skull bones have not fully joined. Normally, this soft spot closes completely between one year and one and a half years of age. In many cultures, mothers recognize that when the soft spot sinks inward, the baby may be in danger. Different beliefs exist to explain this. Some people believe the baby's brain has slipped downward and try to fix it by sucking on the soft spot, pushing up on the roof of the mouth, or holding the baby upside down. These methods do not help because a sunken soft spot is actually a sign of dehydration.

Dehydration means the baby is losing more liquid than he is taking in. This usually happens because of diarrhea, sometimes combined with vomiting. The correct treatment is to give the baby plenty of fluids such as rehydration drink, breast milk, or boiled water. In most cases of diarrhea, medicines are not needed and may even cause harm. If vomiting or diarrhea continues, the underlying cause should be treated appropriately.

If the soft spot is swollen or bulging outward instead of sunken, this may be a sign of meningitis, which is a serious infection. Immediate treatment and medical help are necessary in that case.

The fact that many people use a home remedy does not mean it is safe or effective. It is often difficult to know which remedies truly work. Careful study is needed. Some general guidelines can help identify remedies that are unlikely to work or may be dangerous.

When there are many different remedies for the same illness, it is less likely that any of them are effective. If one truly worked, others would not be needed. For example, many unusual treatments for goiter exist, but none are helpful. The effective prevention and treatment for goiter is using iodized salt.

Remedies that are foul, disgusting, or involve decaying substances are usually not helpful and can be harmful. Some traditional ideas suggest curing diseases by consuming or applying rotten or infected materials. Such practices can cause serious infections and delay proper medical care.

Remedies that involve human or animal waste are especially dangerous. Applying feces or dung to wounds, eyes, burns, or the umbilical cord can cause severe infections, including tetanus. Teas made from waste products do not cure illness and may make a person more sick. These practices should never be used.

Another warning sign is when a remedy closely resembles the illness it claims to cure. In such cases, any benefit usually comes from belief rather than from a true medical effect. For example, using a red object to treat bleeding or using a part of an animal to treat a bite from that animal has no medical value. These remedies may offer comfort if people believe in them, but they do not cure disease. For serious conditions, relying on such remedies can delay life-saving treatment.

Many plants have healing properties. Some modern medicines were originally made from wild herbs. However, not all plants that people believe are curative actually have medical value. Even useful plants can be dangerous if used incorrectly. It is important to learn which medicinal plants in a local area are truly helpful and how to use them safely. Some herbs are poisonous if taken in excessive amounts. In many cases, modern medicines are safer because their dosage can be controlled more precisely.

Angel's Trumpet is a plant whose leaves contain substances that can calm intestinal cramps, stomach pain, and gallbladder pain. The leaves can be ground and soaked in water, and only a small number of drops should be taken at a time. However, this plant is highly poisonous if more than the recommended dose is used, and it should only be given to adults with extreme caution.

Corn silk, which is the fine thread-like part of maize, can be boiled to make a tea that increases urination. This may help reduce swelling in the feet, especially in pregnant women. It is generally safe when prepared properly.

Garlic can be used to help treat pinworms. Crushed garlic mixed with water, juice, or milk can be taken daily for several weeks. Garlic is also used in some traditional treatments for vaginal infections.

Cardon cactus juice can be used to clean wounds when boiled water is not available. The juice may also help stop bleeding by helping blood vessels tighten. A clean piece of cactus can be pressed against a wound to control bleeding temporarily. After bleeding is controlled, the wound should be properly cleaned with soap and boiled water as soon as possible.

Aloe vera is useful for minor burns and wounds. The thick inner gel can reduce pain and itching, promote healing, and help prevent infection. Aloe can also be prepared as a drink to help with stomach ulcers and gastritis, though it has a bitter taste and should be used carefully.

Papaya is rich in vitamins and helps digestion, especially in weak or elderly people who have difficulty digesting meat or other heavy foods. Papaya can also help remove intestinal worms, although modern medicines are usually more effective. The milky substance from unripe papaya or powdered seeds can be used in traditional worm treatments. Papaya is also used to treat pressure sores because it helps soften and remove dead tissue when applied carefully and hygienically.

In some regions, plant-based materials are used to make homemade casts for broken bones. A syrup can be made from certain plant bark and applied to cloth strips, which harden when dry. When applying a cast, the broken bone must first be placed in the correct position. The skin should be protected with soft cloth and padding before applying the hardened strips. The cast should be firm but not too tight and usually should cover the joints above and below the fracture to prevent movement. However, some traditional practices allow limited movement for simple fractures, and research suggests slight movement may sometimes promote healing.

A temporary splint can be made from sturdy materials like cardboard or dried plant stems. After applying a cast, it is important to monitor for swelling. If fingers or toes become cold, pale, or blue, or if the person feels severe tightness, the cast must be removed and replaced with a looser one. A cast should never be placed over an open wound.

Medicinal plants have been used for healing for many years. Some modern medicines are made from wild herbs. However, not all herbs that people use are truly helpful, and some are used incorrectly. It is important to learn which plants in your area are useful and how to use them properly. Some medicinal plants can be very poisonous if taken in large amounts. Because of this, modern medicines are often safer since their doses are easier to control.

Angel’s Trumpet is a plant whose leaves contain substances that help calm intestinal cramps, stomach pain, and gallbladder pain. To prepare it, one or two leaves are ground and soaked in water for a day. Adults may take a small number of drops every few hours. However, this plant is very poisonous if taken in more than the recommended amount.

Corn silk, which is the fine threads from maize, can be made into a tea. This tea increases urine flow and can help reduce swelling in the feet, especially in pregnant women. It is considered safe.

Garlic can help treat pinworms. Crushed garlic cloves are mixed with water, juice, or milk and taken daily for several weeks. Garlic is also used in some cases of vaginal infections.

Cardon cactus juice can be used to clean wounds when clean boiled water is not available. The juice can also help stop bleeding by tightening blood vessels. A clean piece of cactus is pressed against the wound, and after bleeding stops, it can be tied in place for a short time. Later, the wound should be cleaned properly with boiled water and soap.

Aloe vera is useful for minor burns and wounds. The thick gel inside the leaf helps reduce pain and itching, promotes healing, and helps prevent infection. The gel is applied directly to the affected area. Aloe vera can also help in cases of stomach ulcers and gastritis when prepared as a drink from soaked leaves.

Papaya is rich in vitamins and helps digestion. It is especially helpful for older or weak people who have trouble digesting meat or eggs. Papaya can also help remove intestinal worms, though modern medicines work better. The milky juice from green papaya or powdered seeds can be used for this purpose. Papaya can also help treat pressure sores by softening dead tissue, which makes cleaning easier.

Homemade casts can be made from plant-based syrups that dry hard. In some places, plants such as tepeguaje are boiled to make a thick syrup. Cloth strips are dipped in the syrup and wrapped around a properly positioned broken limb. The cast should not be placed directly on the skin. A soft cloth and padding should be used underneath. The cast should cover the joints above and below the break to prevent movement. Fingertips or toes should remain visible to check circulation. If swelling causes the cast to become too tight, it must be removed and replaced. A cast should never be placed over an open wound.

Many people use enemas and laxatives too often. Some believe that fever or diarrhea can be treated by washing out the body with enemas or purges. However, these methods often cause harm, especially to an already sick digestive system.

Enemas and laxatives should never be used in cases of severe stomach pain, suspected appendicitis, gut injury, bullet wounds, or in very weak or sick individuals. They should not be given to babies under two years old. They must not be used in children who have high fever, vomiting, diarrhea, or dehydration because they can make dehydration worse. Frequent use should be avoided.

Simple enemas with warm water may help relieve constipation. In cases of severe dehydration with vomiting, rehydration fluid may be given slowly as an enema.

Strong purges should only be used in cases of poisoning when it is necessary to remove the poison quickly. Otherwise, they are harmful. Mild laxatives may be used in small doses for constipation. Mineral oil may help people with hemorrhoids, but it is not ideal and should not be taken with meals.

The best way to prevent constipation is to drink plenty of water and eat foods rich in natural fiber such as whole grains, bran, cassava, yam, fruits, and vegetables. People who eat natural, unrefined foods suffer less from constipation, hemorrhoids, and gut diseases than those who eat refined foods.

People from different backgrounds explain sickness in different ways. When a baby gets diarrhea, some villagers may believe it happens because the parents did something wrong or because a spirit is angry. A doctor may explain that the cause is an infection. A public health worker may say the problem is due to poor sanitation or unsafe water. A social reformer may believe it is caused by poverty and unfair distribution of resources. A teacher may think it is due to lack of education. Each explanation may be partly correct because sickness often has more than one cause.

To prevent and treat illness successfully, it is important to understand the common diseases in a community and the different factors that cause them. Modern medical science groups diseases into two main types: infectious and non-infectious.

Infectious diseases are illnesses that spread from one person to another. Healthy people must be protected from contact with infected individuals. These diseases are caused by living organisms such as bacteria, viruses, fungi, and parasites.

Non-infectious diseases do not spread from person to person. They are not caused by germs or other living organisms. Antibiotics do not help treat non-infectious diseases.

Non-infectious diseases can be caused by many factors. Some occur when parts of the body wear out or stop working properly, such as heart attacks, strokes, cancer, migraines, cataracts, rheumatism, and epilepsy. Some are caused by harmful substances or outside influences such as allergies, asthma, poisons, snakebites, smoking-related cough, stomach ulcers, or alcoholism. Some result from nutritional deficiencies like malnutrition, anemia, pellagra, night blindness, goiter, and certain liver problems. Some conditions are present at birth, such as cleft lip, crossed eyes, or other deformities. Others begin in the mind and affect thoughts or emotions, including anxiety, paranoia, hysteria, and certain mental delays.

Infectious diseases are caused by harmful organisms that enter and attack the body. Bacteria are tiny living organisms that can cause illnesses such as tuberculosis, tetanus, some types of diarrhea, certain pneumonias, sexually transmitted infections, ear infections, and infected wounds. These infections may spread through air, dirty wounds, contaminated water, sexual contact, or direct touch. Bacterial infections are often treated with specific antibiotics.

Viruses are even smaller than bacteria and cause illnesses such as colds, flu, measles, mumps, chickenpox, polio, viral diarrhea, rabies, warts, and HIV infection. Viruses can spread through air, contact, animal bites, or body fluids. Antibiotics do not work against most viruses. Viral infections are usually treated with supportive care, although some viral diseases can be prevented by vaccination. HIV is treated with antiretroviral medicines.

Fungal infections such as ringworm, athlete’s foot, and jock itch spread through contact with infected skin or clothing. They are treated with antifungal medicines.

Internal parasites live inside the body. Worms and amebas infect the gut and spread through poor hygiene and contaminated food or water. Malaria parasites infect the blood and are spread by mosquito bites. These infections require specific medicines.

External parasites such as lice, fleas, bedbugs, and scabies live on the body and spread through close contact or shared clothing. Good hygiene and specific treatments are used to control them.

Antibiotics are medicines that fight bacterial infections. They do not cure viral infections such as colds or flu. Using antibiotics for viral illnesses is ineffective and may cause harm.

Sometimes different diseases look very similar even though they have different causes and need different treatments. For example, a child who becomes very thin and weak while the belly becomes swollen may have several possible problems. These include malnutrition, heavy roundworm infection, advanced tuberculosis, long-term urinary infection, liver or spleen disease, or even blood cancer such as leukemia. Because the treatments for these illnesses are very different, it is important to identify the exact cause.

An older person with a large, open sore on the ankle that grows slowly may also have different possible conditions. These include poor blood circulation due to varicose veins, diabetes, bone infection, leprosy, skin tuberculosis, or advanced syphilis. Each of these diseases requires a different type of medical treatment.

Many illnesses appear similar at first. By asking careful questions and examining signs closely, it is often possible to find clues that help identify the correct disease. However, diseases do not always show typical signs, and sometimes the symptoms are confusing. In difficult cases, help from a trained health worker or doctor is necessary. Special tests may also be required.

In many communities, people use traditional names for illnesses. These names were created long before germs and modern medicine were understood. As a result, several different diseases that cause similar symptoms may be grouped under one common name. Because doctors may not use these traditional names, people sometimes believe these are special illnesses that doctors do not treat. In reality, most traditional or “folk” diseases are the same diseases recognized by medical science, but with different names.

Some home remedies may help certain illnesses, but others require modern medical treatment, especially serious infections such as pneumonia, typhoid, tuberculosis, or infections after childbirth. When a disease seems serious or unclear, it is important to seek medical help.

One example of a traditional name is a term used for stomach problems that is believed to be caused by a blockage in the gut. In some communities, any illness that causes stomach pain or diarrhea may be given this name. People may believe that a ball of hair or another object blocks the intestine and may use magical treatments. However, the actual causes may include diarrhea with cramps, worm infection, malnutrition with swollen belly, indigestion, stomach ulcer, or in rare cases, true intestinal blockage or appendicitis. The correct approach is to identify and treat the real underlying disease.

Another traditional name refers to side pain in women, usually on one side of the lower abdomen and sometimes spreading to the back. Possible medical causes include urinary tract infection, gas pain or cramps, menstrual pain, appendicitis, infection or tumors of the uterus or ovaries, or even pregnancy outside the womb. Proper examination is necessary to find the true cause.

In some communities, any sudden illness that causes severe distress is given one common name. People may describe congestion of the head, chest, stomach, or whole body. It is often believed to occur when someone breaks food taboos, especially after childbirth, during illness, or while taking medicine. Many foods that are feared are actually harmless, but people avoid them because they fear becoming sick.

Different medical conditions may be grouped under this single name. These include food poisoning from spoiled food, which causes sudden vomiting, diarrhea, cramps, and weakness. Severe allergic reactions can also be included. These may happen after eating certain foods or taking medicines and can cause breathing difficulty, rash, vomiting, and shock. Sudden stomach problems, breathing problems such as asthma or pneumonia, seizures, paralysis, and even heart attacks may also be called by this same name. Each of these illnesses has a different cause and requires different treatment.

Another traditional name refers to a visible pulsing in the upper abdomen. This pulsing is actually the normal heartbeat felt through a large blood vessel. It is more noticeable in very thin or malnourished people. It is often a sign of hunger or malnutrition. The only real treatment is improved nutrition.

There is also a traditional term used for illness believed to be caused by fright, witchcraft, or evil spirits. A person with this condition may appear very nervous, fearful, unable to sleep, lose weight, or behave strangely. In adults, this condition is often related to anxiety or hysteria caused by strong fear or belief. Fear itself can cause physical weakness and worsening symptoms. In children, frightening dreams or high fever may cause strange behavior. Malnutrition, tetanus, or meningitis may sometimes be mistaken for this condition.

Treatment depends on the cause. If a real illness is present, that illness should be treated. If fear is the main cause, reassurance and emotional support are important. In cases where a frightened person breathes very fast and deeply, hyperventilation may occur. Signs include fast breathing, pounding heart, tingling in hands or face, and muscle cramps. The person should be kept calm and asked to breathe slowly into a paper bag for a few minutes. Reassurance helps recovery.

Misunderstandings often happen because villagers and medical workers use the same word for different conditions. For example, a word commonly used by villagers for severe infected wounds or gangrene is also the medical word for cancer. In medical science, cancer is not an infection but an abnormal growth of cells that forms a lump. It can occur in the skin, breast, womb, ovaries, or other parts of the body. A hard, painless, slowly growing lump may be cancer and requires medical attention.

Another traditional word is used for any spreading skin sore. However, in medical terms, this word refers specifically to Hansen’s disease. Many common skin infections, insect bite sores, chronic ulcers, and even skin cancer may be wrongly called by this name. Proper examination is necessary to distinguish them.

Fever is another example of confusion. In many places, different serious diseases that cause high body temperature are all called simply fever. However, these diseases have different patterns and signs.

Malaria usually begins with weakness, chills, and fever that comes and goes. The person may shiver as temperature rises and sweat as it falls. The fever may return every few days.

Typhoid fever begins gradually like a cold. The temperature rises a little more each day. The pulse may be slower than expected. The person may have diarrhea, dehydration, and mental confusion.

Typhus is similar to typhoid but includes a rash with small bruises.

Hepatitis causes loss of appetite, nausea, yellowing of the skin and eyes, dark urine, pale stools, mild fever, and weakness.

Pneumonia causes fast breathing, high fever that rises quickly, cough with colored mucus, and chest pain.

Rheumatic fever usually affects children or teenagers. It causes high fever and joint pain, often after a sore throat. It may also affect the heart and cause uncontrolled movements.

Brucellosis begins slowly with tiredness and bone pain. Fever and sweating often occur at night. The fever may disappear and return repeatedly for months.

Childbirth fever begins after delivery with rising temperature, foul-smelling discharge, and pain.

All these diseases can be dangerous. Long-lasting fever or night sweats may also be caused by HIV infection. Because symptoms may overlap, medical help should be sought whenever possible.

To examine a sick person properly, first ask important questions and then examine the person carefully. Look for signs and symptoms that help you understand how serious the illness is and what kind of disease it may be. Always examine the person in good light, preferably sunlight, and never in a dark room.

When someone is sick, there are basic things to ask and observe. Symptoms are what the person feels and tells you. Signs are what you notice during examination. In babies and people who cannot speak, signs are especially important. Write down all your findings so they can be shared with a health worker if needed.

Begin by asking what problem is troubling the person most right now. Ask what makes the problem better or worse. Ask when and how the illness began. Find out if the person has had the same problem before or if someone else in the family or neighborhood has had it. Continue asking questions to understand the illness clearly.

If the person has pain, ask where it hurts and ask them to point to the exact place. Ask if the pain is constant or comes and goes. Ask what kind of pain it is, such as sharp, dull, or burning, and whether it affects sleep. If the sick person is a baby, look for signs of pain through movements and crying. For example, a baby with ear pain may pull at the ear or rub the side of the head.

Before touching the person, observe their general condition. Notice how sick or weak they appear, how they move, how they breathe, and whether they are mentally alert. Look for signs of dehydration and shock. Check whether the person looks well nourished or poorly nourished. Slow weight loss over a long time may suggest a chronic illness.

Observe the color of the skin and eyes. Look at lighter areas like the palms, soles, fingernails, and inside the lips and eyelids. Pale lips and eyelids may suggest anemia. Very light skin color may also occur in long-term illness or severe malnutrition. Darkening of the skin may suggest starvation. Bluish lips or fingernails may indicate serious breathing or heart problems. A gray-white color with cool, moist skin often indicates shock. Yellow skin and eyes suggest jaundice, which may be due to liver or gallbladder disease. Yellow color can also occur in newborns or in children with sickle cell disease. Looking at the skin from the side with light shining across it may help detect early measles rash in a child with fever.

Taking the temperature is important, even if the person does not feel hot. If the person is very sick, check the temperature at least four times a day and record it. Understanding how the fever starts, how high it rises, how long it lasts, and how it goes away can help identify the illness. Not all fevers are malaria. Mild fever often occurs in common colds and viral infections. Typhoid fever usually rises gradually over five days and does not improve with malaria medicine. Tuberculosis may cause mild afternoon fever and night sweating.

If no thermometer is available, you can compare the temperature by placing one hand on the sick person's forehead and the other on a healthy person's forehead.

To use a thermometer, clean it well with soap, water, or alcohol. Shake it until it reads below 36°C. Place it under the tongue with the mouth closed, in the armpit if there is risk of biting, or carefully in the anus of a small child after applying grease. Leave it in place for three to four minutes, then read it and wash it again. Armpit readings are slightly lower than mouth readings, and rectal readings are slightly higher. In newborn babies, unusually high or low temperature may indicate serious infection.

Observe breathing carefully. Notice whether breaths are deep or shallow, how fast they occur, and whether breathing seems difficult. Watch if both sides of the chest move equally. Count breaths per minute when the person is resting. Adults and older children normally breathe 12 to 20 times per minute. Children may breathe up to 30 times per minute, and babies up to 40 times per minute.

Rapid breathing may occur with high fever or lung illness. More than 30 shallow breaths per minute in an adult may suggest pneumonia. Sixty breaths per minute in a newborn may also suggest pneumonia. Listen to the breathing sounds. A whistling sound may indicate asthma. Gurgling or snoring sounds in an unconscious person may mean something is blocking the throat. If the skin between the ribs or at the neck pulls inward during breathing, it suggests difficulty in getting air and may be caused by pneumonia, asthma, bronchitis, or a blocked airway.

If there is a cough, ask if it disturbs sleep. Ask whether mucus is produced, how much, its color, and whether there is blood.

Check the pulse by placing your fingers on the wrist without using your thumb. If you cannot feel it there, check the neck beside the voice box or listen to the chest. At rest, normal pulse is 60 to 80 beats per minute in adults, 80 to 100 in children, and 100 to 140 in babies. The pulse increases with exercise, fear, or fever. Usually, for each one-degree rise in temperature, the pulse increases by about 20 beats per minute.

Record pulse, temperature, and breathing rate in very sick persons. A weak and rapid pulse may indicate shock. Very fast, very slow, or irregular pulse may indicate heart problems. A relatively slow pulse with high fever may suggest typhoid.

Examine the eyes. Look at the white part for redness or yellow color. Ask about vision changes. Ask the person to move their eyes slowly in all directions. Jerking or uneven movement may suggest brain damage. Check the pupils, the black center of the eyes. Very large pupils may indicate shock. Very large or very small pupils may indicate poisoning or drug effects. A white glow may indicate cataract or cancer.

Compare both pupils carefully. A large difference in size is usually an emergency. If the eye with the larger pupil is very painful and causes vomiting, glaucoma may be the cause. If the eye with the smaller pupil is very painful, it may be iritis. In unconscious persons or those with head injury, unequal pupils may indicate brain damage or stroke.

Always compare the pupils in unconscious persons or those with head injury.

When examining the ears, always check for signs of pain or infection, especially in a child who has fever or a cold. A baby who cries often or pulls at the ear may have an ear infection. Gently pull the ear. If this increases pain, the infection is likely in the ear canal. Look inside the ear for redness or pus using a small flashlight. Never insert sticks, wires, or hard objects into the ear. Also check hearing by rubbing your fingers near each ear to see if the person can hear equally on both sides.

Examine the throat and mouth using sunlight or a flashlight. Ask the person to open the mouth and say “ah” or gently press the tongue down with a spoon handle. Check if the throat is red. Look at the tonsils at the back of the throat to see if they are swollen or have white spots of pus. Examine the mouth for sores, swollen gums, painful tongue, rotten teeth, abscesses, or other problems.

Check the nose to see if it is runny or blocked. Observe how a baby breathes through the nose. Shine a light inside the nose and look for mucus, pus, blood, redness, swelling, or bad smell. Also look for signs of sinus infection or allergy.

Examine the entire skin of the sick person, even if the illness appears mild. Babies and children should be fully undressed for proper examination. Look carefully for sores, wounds, splinters, rashes, welts, spots, patches, unusual marks, redness, heat, swelling, or pain that may indicate infection. Check for swelling or puffiness anywhere on the body. Feel for swollen lymph nodes in the neck, armpits, or groin. Look for abnormal lumps or masses. Notice thinning hair, hair loss, loss of shine, or change in hair color. Loss of eyebrows may suggest certain chronic diseases.

In children, carefully check between the buttocks, around the genital area, between fingers and toes, behind the ears, and in the scalp. Look for lice, scabies, ringworm, rashes, and sores.

If a person has abdominal pain, first ask them to point with one finger to the exact location of pain. Ask whether the pain is constant or comes and goes like cramps. Examine the abdomen by first looking for swelling or lumps. Then gently press different areas of the abdomen, starting away from the painful area and moving toward it, to find where it hurts most.

Check whether the abdomen feels soft or hard. A very hard abdomen may indicate a serious condition such as appendicitis or peritonitis. If such conditions are suspected, test for rebound tenderness. Feel carefully for abnormal lumps or hardened areas.

If the person has constant stomach pain, nausea, and has not passed stool, listen to the abdomen using your ear or a stethoscope. Listen for intestinal gurgling sounds. If no sounds are heard after about two minutes, this is a serious danger sign.

Pain location in the abdomen can suggest specific problems. Pain in the upper middle abdomen may indicate an ulcer. Pain from appendicitis often starts near the middle of the abdomen and later moves to the lower right side. Gallbladder pain is usually on the right side and may spread to the back. Liver pain is on the right upper side and may spread to the chest. Urinary system problems may cause mid or lower back pain that spreads toward the lower abdomen. In women, inflammation, tumor of the ovaries, or ectopic pregnancy may cause pain on one or both sides of the lower abdomen, sometimes spreading to the back.

When examining muscles and nerves, pay attention if the person complains of numbness, weakness, or loss of control in any part of the body. Observe how the person walks and moves. Ask them to stand, sit, or lie straight, and compare both sides of the body carefully to look for differences.

Examine the face by asking the person to smile, frown, open the eyes wide, and close them tightly. Look for drooping or weakness on one side of the face. If facial weakness starts suddenly, it may be due to head injury, stroke, or Bell’s palsy. If it develops slowly, it could be a brain tumor, and medical advice is needed. Also check eye movements, pupil size, and vision.

Look at the arms and legs for muscle loss. Compare the size and thickness of both sides. Watch how the person walks and moves. If weakness or muscle loss affects the whole body, it may be due to malnutrition or a long-term illness such as tuberculosis.

Test muscle strength by asking the person to squeeze your fingers and comparing the strength of both hands. Ask them to push and pull against your hand with their feet. Have them hold their arms straight out and turn their hands up and down. Ask them to lie down and lift one leg at a time. Observe for weakness or trembling.

If muscle weakness is uneven or worse on one side, in children consider polio. In adults, think about back problems, head injury, or stroke.

Check for stiffness or tightness of muscles. A stiff jaw that will not open may indicate tetanus, severe throat infection, tooth infection, or jaw dislocation if there was recent injury. If the neck or back is stiff and bent backward in a very sick child, suspect meningitis. If the head cannot bend forward toward the chest, meningitis is likely.

If a child always has stiff muscles and makes strange or jerky movements, the condition may be spasticity. If jerky movements occur suddenly with loss of consciousness, they may be seizures. Repeated seizures suggest epilepsy. Seizures during illness may be caused by high fever, dehydration, tetanus, or meningitis.

To check for loss of sensation, ask the person to close their eyes. Lightly touch or gently prick different areas of the skin and ask them to say when they feel it. Loss of feeling in specific patches may suggest leprosy. Loss of feeling in both hands or feet may be due to diabetes or leprosy. Loss of feeling on only one side may result from a back problem or injury.

Sickness weakens the body, so proper care is necessary for recovery. Medicines are not always required, but good care is always important.

A sick person should rest in a quiet and comfortable place with fresh air and good light. The person should not become too hot or too cold. If the weather is cold or the person feels chilled, cover them with a blanket. If the weather is hot or the person has a fever, avoid covering them too much.

In most illnesses, especially when there is fever or diarrhea, the person should drink plenty of liquids such as water, tea, juices, or broth.

Cleanliness is very important. The sick person should be bathed daily. If too weak to get out of bed, wash them gently with lukewarm water using a cloth. Clothes, sheets, and blankets must be kept clean. Food crumbs and dirt should not remain in the bed.

If the sick person feels hungry, allow them to eat. Most illnesses do not require special diets. The person should eat nourishing food and drink plenty of liquids. If very weak, give small amounts of nourishing food many times a day. Foods can be mashed or made into soups or juices if chewing is difficult. Energy-rich foods such as rice, wheat, oatmeal, potato, or cassava porridges are helpful. Adding a small amount of sugar or vegetable oil increases energy. Sweetened drinks may help if the person eats little.

Some illnesses require special diets, such as anemia, stomach ulcers, appendicitis, diabetes, heart disease, gallbladder problems, and high blood pressure. In conditions like appendicitis or intestinal obstruction, no food should be given.

For a very ill person, liquid intake is extremely important. If they can drink only small amounts, give small sips frequently. Measure how much liquid they drink daily. An adult should drink about two liters or more per day and urinate at least three or four times daily. If urination decreases or signs of dehydration appear, encourage more fluids. Nutritious liquids with a little salt are helpful. If needed, give oral rehydration solution. Intravenous fluids may be required in severe dehydration, but frequent small sips can often prevent this.

If solid food cannot be eaten, provide soups, milk, juices, and broths. Porridges should be combined with body-building foods like egg, beans, meat, fish, or chicken. Small frequent meals are better than large ones.

A seriously ill person must be kept very clean. Bathe daily with warm water. Change bed sheets daily or whenever dirty. Clothes and bedding soiled by an infectious person should be washed in hot soapy water or with bleach to kill germs.

A very weak person who cannot move alone should be helped to change position in bed several times a day to prevent bed sores. Frequent position change also helps prevent pneumonia, especially in people who remain in bed for long periods. If a bedridden person develops fever, cough, and fast shallow breathing, pneumonia may be present.

Caregivers must watch for changes in condition. Vital signs such as temperature, pulse, and breathing rate should be recorded four times daily. Also record fluid intake, urination, and bowel movements. These records help health workers assess the patient's condition.

Certain signs indicate dangerous illness and require immediate medical help. These include heavy bleeding, coughing blood, sudden blue lips and nails, severe breathing difficulty, unconsciousness, fainting on standing, inability to urinate or drink for a day, severe vomiting or diarrhea lasting more than a day, black tar-like stool or bloody vomit, severe continuous abdominal pain, strong pain lasting more than three days, stiff neck with arched back, repeated convulsions with fever, high fever above 39°C lasting more than four to five days, long-term weight loss, blood in urine, sores or lumps that continue to grow, and serious problems during pregnancy such as bleeding, swelling with vision problems, long delay after water breaks, or severe bleeding.

Medical help should be sought as soon as signs of dangerous illness appear. Do not wait until the person becomes too weak to be moved to a health center or hospital. Early treatment can prevent serious complications.

If moving the sick or injured person could worsen their condition, try to bring a health worker to the patient. However, in emergencies where special treatment or surgery may be required, such as suspected appendicitis, do not wait. Take the person to a hospital immediately.

When carrying a person on a stretcher, make sure they are comfortable and secure to prevent falling. If there are broken bones, they should be splinted before moving the person. Protect the patient from strong sun by creating shade while allowing fresh air to circulate.

For proper treatment, a health worker should ideally examine the sick person directly. If the patient cannot travel, request the health worker to come. If this is not possible, send a responsible adult who understands the details of the illness. Before seeking help, carefully examine the patient and write down all findings, including symptoms, duration, fever, pain, changes in skin, breathing, urine, stools, and any dangerous signs. Also record any medicines being taken and any history of allergic reactions.

Most illnesses do not require medicines. The body has natural defenses that fight disease. In many cases, recovery depends more on rest, cleanliness, proper nutrition, and adequate fluids than on medication. Even when medicine is necessary, it only supports the body's natural healing processes.

Water is essential for life and health. A large portion of the human body is made up of water. Proper use of water can prevent many illnesses and reduce death, especially in children.

Safe drinking water is critical in preventing diarrhea and intestinal infections. Wells and springs should be protected from contamination by animals and dirt. Drainage around water sources should prevent dirty water from flowing into them. In areas where water may be unsafe, it should be boiled or filtered before drinking or preparing food. This is especially important for babies. Baby bottles and utensils should be boiled. If boiling bottles is not possible, using a clean cup and spoon is safer.

Handwashing with soap and water after bowel movements and before handling food is very important in preventing disease.

A common cause of death in children with diarrhea is dehydration. Giving plenty of fluids, especially water mixed with sugar and salt or cereal, can prevent or treat dehydration. In many cases, giving enough fluids is more important than giving medicine.

Water can be used effectively in both prevention and treatment of many conditions. Regular bathing helps prevent skin infections. Washing wounds thoroughly with soap and clean water helps prevent infection and tetanus. Cold water soaks during the first day help strains and sprains, followed by warm soaks later. Cold compresses can relieve itching and burning skin. Minor burns should be placed immediately in cold water. Gargling with warm salt water helps sore throat. If harmful substances enter the eye, rinse immediately with cool water for at least thirty minutes. Sniffing salt water can relieve a blocked nose. Drinking plenty of water helps constipation. Applying ice early may help cold sores.

In many of these conditions, proper use of water may reduce or eliminate the need for medicines. Medicines should be used only when absolutely necessary.

Right and Wrong Uses of Modern Medicines

Some medicines sold in pharmacies or village stores are very useful, but many are not necessary. Out of the thousands of medicines sold in most countries, only a small number are truly essential. Many medicines are used incorrectly, and even good medicines can cause harm if they are not used properly. For medicine to help, it must be used correctly and only when needed.

Many doctors and health workers prescribe more medicines than necessary. This overuse leads to avoidable sickness and even death. Some medicines are more dangerous than others, yet people sometimes use very dangerous medicines for mild illnesses. A dangerous medicine should never be used for a mild sickness. Medicines can kill if they are misused.

Medicines should be used only when necessary. It is important to know the correct way to use a medicine and understand its precautions. The right dose must always be given. If a medicine does not help or causes problems, it should be stopped. When unsure, advice from a trained health worker should be sought. Patients should tell doctors they want medicine only if it is truly needed. This saves money and protects health.

One of the most dangerous mistakes is using powerful medicines for mild conditions. Chloramphenicol is sometimes used for simple diarrhea or mild infections, but it is very risky and should only be used for severe diseases like typhoid. It must never be given to newborn babies.

Some medicines such as oxytocin, ergonovine, and misoprostol are sometimes used to speed up childbirth. This is extremely dangerous and can kill the mother or baby. These medicines should only be used to control bleeding after childbirth.

Many people believe injections are better than medicines taken by mouth, but this is not true. Often, medicine taken by mouth works just as well or better. Injections can be more dangerous and may cause serious complications, including paralysis in children with certain infections. Injections should be used only when absolutely necessary.

Penicillin works only against specific bacterial infections. It should not be used for sprains, bruises, pain, or fever. Injuries that do not break the skin usually do not need antibiotics. Antibiotics do not cure colds. Penicillin can also cause serious allergic reactions in some people.

Kanamycin and gentamicin are antibiotics that can cause permanent hearing loss in babies if overused. They should be given to babies only for life-threatening infections. In many cases, safer antibiotics can be used instead.

Certain anti-diarrhea medicines containing hydroxyquinolines can cause severe side effects such as paralysis, blindness, and even death. They are banned in many countries but are still sold in some places. These medicines should not be used.

Cortisone and other steroid medicines are powerful drugs used for severe asthma, arthritis, or allergic reactions. However, they are often misused for minor pain because they give quick relief. Steroids can cause serious side effects, especially when used in high doses or for long periods. They weaken the body's defense against infection, worsen tuberculosis, cause stomach bleeding, and weaken bones.

Anabolic steroids are sometimes given to children to help them grow or gain weight. Although they may cause faster growth at first, they eventually stop growth earlier and result in shorter adult height. They also cause serious side effects, such as permanent facial hair growth in girls. Children should not be given growth tonics. Proper nutrition is safer and more effective.

Certain medicines used for arthritis can cause dangerous blood disorders and damage to the stomach, liver, and kidneys. Safer and cheaper medicines are available for pain and fever.

Vitamin B12 and liver extract do not usually help anemia except in rare cases. Injecting them can cause risks and should only be done after proper testing and specialist advice. Injectable iron is also risky. Iron tablets are safer and work just as well for most cases of anemia.

Vitamins should generally not be injected because injections are more dangerous and not more effective than pills. Many vitamin syrups and tonics are unnecessary and a waste of money. A balanced diet with beans, eggs, meat, fruits, vegetables, and whole grains provides essential vitamins and nutrients. Good food usually helps more than supplements.

Combination medicines contain two or more drugs in one pill. They are often less effective, more expensive, and sometimes harmful. Only necessary medicines should be used. Some combination medicines for HIV are useful because they make treatment easier. However, many common combination products, such as cough mixtures, antibiotics mixed with anti-diarrhea medicines, or multiple painkillers combined together, should be avoided.

Injecting calcium into a vein is extremely dangerous and can cause death if done incorrectly. Calcium injections into muscles can cause severe infections. In areas where people eat foods rich in natural calcium, supplements or injections are usually unnecessary.

Intravenous fluids are often wrongly believed to strengthen weak or anemic people. These fluids are mostly water with salt or sugar and do not improve anemia or strength. They can also cause serious infections if not given properly. Intravenous fluids should only be used when a person cannot drink or is severely dehydrated. If a person can swallow, oral rehydration with water, sugar, and salt is safer and just as effective. Nutritious food strengthens the body better than intravenous fluids.

When Medicine Should Not Be Taken

Many people believe they should avoid certain foods while taking medicines. In reality, medicines do not become harmful simply because they are taken with specific foods such as pork, chili, guava, or oranges. However, greasy or very spicy foods can worsen stomach problems, whether or not a person is taking medicine. Some medicines can cause dangerous reactions if taken with alcohol, so alcohol should be avoided when using those drugs.

There are situations when certain medicines should clearly not be used. Pregnant women and women who are breastfeeding should avoid all medicines that are not absolutely necessary. However, small amounts of vitamins or iron tablets are generally safe. Women with HIV who are pregnant or breastfeeding should take appropriate medicines to prevent passing the infection to the baby.

Newborn babies require special care when giving medicines. Whenever possible, medical advice should be sought before giving any medicine to a newborn. Extra care must be taken to avoid giving too much.

Anyone who has ever had an allergic reaction, such as hives or itching, after taking a medicine like penicillin, ampicillin, or sulfonamides, should never take that medicine again. A second exposure can cause a much more dangerous reaction.

People who have stomach ulcers or frequent heartburn should avoid medicines containing aspirin. Most painkillers and all steroid medicines can worsen ulcers and acid indigestion. Acetaminophen is a pain reliever that does not usually irritate the stomach.

Some medicines are dangerous for people who have certain illnesses. For example, a person with hepatitis has a damaged liver and should avoid antibiotics or other strong medicines because these drugs can further harm the liver and poison the body.

People who are dehydrated or have kidney disease must be especially careful with medicines. Medicines that can build up in the body should not be given repeatedly unless the person is urinating normally. For example, a dehydrated child with high fever should not receive repeated doses of acetaminophen or aspirin until normal urination begins. Sulfa medicines should never be given to someone who is dehydrated.

Antibiotics: What They Are and How to Use Them

Antibiotics are important medicines used to fight infections caused by bacteria. They do not work against viral infections such as the common cold. Different antibiotics act in different ways and are effective against specific types of bacteria. Examples include penicillin, tetracycline, streptomycin, chloramphenicol, erythromycin, sulfonamides, and cephalosporins.

Each antibiotic may be sold under several brand names, which can cause confusion. It is important to read the label carefully to know the generic name of the medicine and to understand which group it belongs to. Some antibiotics are more dangerous than others and should only be used for severe infections.

Antibiotics must never be used unless the person understands what infection they treat, the correct dose, and the precautions needed. They should only be used for infections they are known to cure. The correct dose depends on the illness and the age or weight of the patient. Taking too much or too little can reduce effectiveness and increase risks.

Injections of antibiotics should be avoided when oral medicines will work just as well. Injections carry additional risks and should be used only when absolutely necessary. Antibiotics should be continued until the infection is completely cured or for at least two days after symptoms such as fever disappear. Some illnesses, like tuberculosis, require treatment for many months even after the patient feels better.

If an antibiotic causes rash, itching, difficulty breathing, or any serious reaction, it must be stopped immediately and never used again. Antibiotics should be used only when clearly needed, because overuse causes bacteria to become resistant, meaning the medicine will no longer work.

Certain precautions apply to specific antibiotics. Before injecting penicillin or ampicillin, emergency treatment for allergic reactions should be available. People allergic to penicillin should use alternative antibiotics such as erythromycin or certain sulfa drugs.

Broad-spectrum antibiotics, which kill many types of bacteria, should not be used when a narrow-spectrum antibiotic would work. Overuse of broad-spectrum drugs increases resistance and side effects.

Chloramphenicol is a dangerous antibiotic and should be used only for severe, life-threatening illnesses such as typhoid. It must never be used for mild illnesses and should not be given to newborn babies except in rare situations.

Tetracycline and chloramphenicol should not be injected because they work well when taken by mouth and injections increase risk. Tetracycline should not be given to pregnant women or to children under eight years of age because it can damage developing teeth and bones.

Streptomycin and related medicines should generally be used only for tuberculosis and always combined with other anti-tuberculosis medicines. These drugs are toxic and should not be used for mild infections such as colds or flu. Medicines in this group, including kanamycin and gentamicin, can be harmful and should be reserved for serious infections.

Eating yogurt or curdled milk can help restore healthy bacteria in the body that are destroyed by antibiotics.

If an antibiotic does not seem to help after one or two days, several reasons are possible. The illness may not be correctly diagnosed and the wrong medicine may be used. The dose may be incorrect. The bacteria may have become resistant to that antibiotic. In serious or worsening cases, medical help should be sought.

Antibiotics do not help the common cold. Many children have been harmed by unnecessary use of antibiotics for mild illnesses. Rest, good food, and fluids are often safer and more effective for minor infections.

Importance of Limited Use of Antibiotics

All medicines should be used carefully and only when necessary, but this is especially important with antibiotics. Antibiotics can harm the body as well as kill bacteria. They may cause poisoning or allergic reactions, and many people die each year from taking antibiotics they do not need.

Antibiotics also disturb the natural balance of bacteria in the body. Not all bacteria are harmful. Some are necessary for normal body functions. When antibiotics kill helpful bacteria, other organisms such as fungi can grow too much. Babies who take antibiotics may develop fungal infections in the mouth or on the skin. People who take broad-spectrum antibiotics like ampicillin for several days may develop diarrhea because the medicine destroys helpful bacteria needed for digestion.

The most serious problem caused by overuse of antibiotics is resistance. When bacteria are repeatedly exposed to the same antibiotic, they can become stronger and no longer respond to that medicine. This is called resistance. As a result, diseases that were once easy to treat, such as typhoid, are becoming harder to cure. In some areas, typhoid no longer responds to chloramphenicol because it has been used too often for mild infections. Worldwide, many serious diseases are becoming resistant to antibiotics mainly because these medicines are overused for minor illnesses.

Most minor infections do not require antibiotics. Minor skin infections usually improve with soap, water, hot compresses, and simple treatments. Mild respiratory infections often get better with rest, fluids, and nutritious food. Most diarrhea does not require antibiotics and may even worsen with them. The most important treatment for diarrhea is drinking plenty of fluids and continuing to eat as soon as possible.

How to Measure and Give Medicine

Medicines are commonly measured in grams and milligrams. One gram equals one thousand milligrams. It is important to understand these measurements to avoid giving too much or too little medicine.

For example, an adult aspirin tablet usually contains 300 milligrams of aspirin, while a baby aspirin contains 75 milligrams. This means one adult tablet contains four times as much medicine as a baby tablet. If necessary, an adult tablet can be divided into four equal pieces to give the same dose as one baby tablet.

Many medicines, especially antibiotics, are available in different strengths. It is important to read the label carefully and check how many milligrams each tablet or capsule contains. If the prescribed dose is 250 milligrams and only 50 milligram capsules are available, several capsules must be taken to equal the correct dose.

Some medicines such as penicillin are measured in units instead of milligrams. It is important to understand the conversion between units and milligrams to give the correct dose.

Liquid medicines such as syrups and suspensions are measured in milliliters. One teaspoon equals five milliliters, and one tablespoon equals fifteen milliliters. Since household teaspoons vary in size, it is safest to use a proper measuring spoon that holds exactly five milliliters.

Medicines for children often come in liquid form, but these are usually more expensive than tablets or capsules. To save money, tablets can sometimes be crushed or capsules opened and mixed with cooled boiled water and sugar to make a liquid. However, great care must be taken to measure the correct dose. Honey should not be given to babies under one year of age because it can rarely cause serious illness.

To prevent choking, medicines should never be given to a child who is lying flat, unconscious, asleep, or having seizures. The child should be sitting upright or with the head slightly forward.

If only the adult dose is known, children should receive smaller amounts depending on age or weight. In general, children between eight and thirteen years may receive about half the adult dose. Children between four and seven years may receive about one quarter of the adult dose. Children between one and three years may receive about one eighth of the adult dose. Infants under one year should usually receive the dose recommended for a one-year-old, but medical advice should be sought whenever possible.

Medicines should be taken at regular intervals as directed. If a medicine is prescribed every eight hours, it should be taken three times a day, such as morning, afternoon, and night. If prescribed every six hours, it should be taken four times a day. If every four hours, it should be taken six times daily with roughly equal spacing.

When giving medicine to someone else, it is important to clearly explain the instructions and ask the person to repeat them to ensure understanding. For people who cannot read, simple drawings or symbols can help remind them when and how much medicine to take.

When You Give Medicines to Anyone

Whenever you give medicine to someone, always write important information on the note that goes with the medicine, even if the person cannot read. Include the person's name, the name of the medicine, what it is used for, and the exact dosage. This information can be written together with simple drawings that show how much to take. It is also wise to keep a record of this same information for yourself. If possible, complete a full patient report.

Taking Medicines on a Full or Empty Stomach

Some medicines work best when taken on an empty stomach, which means about one hour before eating. Others are less likely to cause stomach pain or heartburn if taken with food or right after a meal.

Penicillin, ampicillin, and tetracycline should be taken one hour before meals. Milk should not be consumed one hour before or after taking tetracycline because it reduces its effectiveness.

Aspirin, medicines that contain aspirin, iron tablets, vitamins, and erythromycin are better taken with food or soon after eating, or with plenty of water.

Antacids work best when taken on an empty stomach, usually one or two hours after meals and at bedtime.

It is best to take medicines while sitting or standing upright. Drinking a full glass of water with each dose is helpful. If taking sulfa medicines, it is especially important to drink plenty of water to protect the kidneys.

Instructions and Precautions for Injections

Injections are rarely necessary. Most illnesses can be treated as well or better with medicines taken by mouth. Many people become ill or even die each year because of unnecessary injections. Injections should be used only when absolutely necessary and usually only by trained health workers.

Medicines should be injected only when they are not available in oral form, when the patient cannot swallow or is vomiting repeatedly, or in certain serious emergencies.

If a doctor prescribes injections, it is reasonable to ask whether medicine by mouth would work instead, especially in places where trained persons are not available to give injections safely. Injections of vitamins, liver extract, or vitamin B12 should not be accepted unless proper blood testing has been done.

In certain emergencies, injections may be life-saving. These include severe pneumonia, serious infections after childbirth, gangrene, tetanus, appendicitis, peritonitis, severe belly wounds, poisonous snakebite, scorpion stings in children, meningitis, severe allergic reactions, and uncontrolled vomiting. In chronic illnesses such as tuberculosis, syphilis, or gonorrhea, injections may be required but are rarely emergencies and should be handled by trained professionals.

Injections should never be given for mild illnesses such as colds or flu. Never inject a medicine that is not recommended for the illness. Never inject with an unsterilized needle, and never inject unless you understand the proper precautions.

Certain medicines are generally better not injected. Vitamins, liver extract, vitamin B12, and iron injections are rarely better than pills and can cause dangerous reactions. Calcium injections can be extremely dangerous and should only be given by trained professionals. Penicillin is usually safer when taken by mouth and should only be injected for serious infections. Penicillin combined with streptomycin should not be used for colds or flu. Chloramphenicol and tetracycline are safer and equally effective when taken orally. Intravenous solutions and medicines should only be given by well-trained professionals and only in severe cases such as serious dehydration.

Risks and Precautions of Injections

The two main risks of injections are infection from unclean equipment and allergic or poisonous reactions to the medicine.

To prevent infection, always boil the needle and syringe before use and avoid touching the sterile needle. Never use the same needle for more than one person without boiling it again. Always wash hands thoroughly before preparing or giving an injection.

Some medicines, especially penicillins and certain antitoxins made from horse serum, can cause severe allergic shock shortly after injection. The risk is higher in people who have had allergic reactions to similar medicines in the past.

Before giving such injections, emergency medicines like epinephrine and antihistamines should be available. Always ask if the person has had allergic reactions before. After giving an injection, stay with the person for at least thirty minutes to watch for signs of allergic shock such as pale and sweaty skin, weak rapid pulse, difficulty breathing, or loss of consciousness. If these occur, epinephrine must be given immediately and the person treated for shock.

Any signs such as rash, itching, swelling, breathing difficulty, dizziness, vision problems, ringing in the ears, severe back pain, or difficulty urinating after an injection mean that the medicine should never be given again.

To avoid serious harm, injections should only be given when absolutely necessary, with clean equipment, correct medicine, correct technique, and proper monitoring afterward.

Avoiding Serious Reactions to Penicillin

Before giving a penicillin injection, always ask the person if they have ever had a rash, itching, swelling, or trouble breathing after receiving penicillin in the past. If the answer is yes, do not use penicillin, ampicillin, or amoxicillin. Instead, choose another antibiotic such as erythromycin or a sulfonamide.

Before giving the injection, make sure emergency medicine like epinephrine (adrenaline) is available. After giving the injection, stay with the person for at least thirty minutes to watch for signs of a severe allergic reaction.

If the person suddenly becomes very pale, has a very fast heartbeat, difficulty breathing, or begins to faint, inject epinephrine immediately into a muscle or just under the skin. Use half an ampule for adults and a quarter ampule for small children. If needed, repeat after ten minutes.

Preparing a Syringe for Injection

Before preparing a syringe, wash your hands with soap and water. Take the syringe apart and boil the syringe and needle for twenty minutes. Pour out the boiled water carefully without touching the sterile equipment. Put the syringe and needle back together, touching only the base of the needle and the top of the plunger.

Clean the ampule of distilled water and break off the top carefully. Fill the syringe with distilled water without letting the needle touch the outside of the ampule. If using a bottle with powdered medicine, clean the rubber top with a clean cloth dipped in alcohol or boiled water. Inject the distilled water into the bottle and shake it until the medicine dissolves completely. Fill the syringe again with the prepared medicine and remove all air from the syringe.

Be extremely careful not to touch the needle with anything, including your fingers or cotton. If the needle touches anything unclean, boil it again before using it.

Where to Give an Injection

Before giving an injection, wash your hands. The preferred place for an intramuscular injection is the upper outer part of the buttock. This area reduces the risk of injuring important nerves or blood vessels.

How to Give an Injection

Clean the skin with soap and water or alcohol. If alcohol is used, allow it to dry before injecting to reduce pain. Insert the needle straight in with one quick movement to lessen discomfort. Before injecting the medicine, gently pull back on the plunger. If blood enters the syringe, remove the needle and insert it in a different place. If no blood appears, inject the medicine slowly. After finishing, remove the needle and clean the skin again.

Immediately after use, rinse the syringe and needle. Push clean water through the needle, then take the syringe apart and wash it. Boil it again before the next use.

How Injections Can Harm Children

When used correctly, some injections such as vaccines protect children from disease and disability. However, if injections are given with unsterilized equipment, serious infections can occur. Dirty needles and syringes can spread HIV, hepatitis, tetanus, and other dangerous diseases. They can also cause severe infections that lead to paralysis or death.

Never use the same needle or syringe on more than one person unless it has been properly sterilized. Some injected medicines can also cause allergic reactions, poisoning, deafness, or other harmful effects. For example, hormone injections given to pregnant women to speed childbirth can be dangerous and may cause brain damage or death in the baby.

Cleaning and Sterilizing Equipment

Many infectious diseases spread through unsterilized instruments, including syringes, needles, and tools used for ear piercing, tattoos, acupuncture, or circumcision. Whenever the skin is cut or pierced, sterile equipment must be used.

Equipment can be sterilized by boiling it for twenty minutes. If no clock is available, adding a few grains of rice to the water can help; when the rice is cooked, the equipment has boiled long enough. Equipment can also be sterilized by pressure steaming in a pressure cooker or autoclave for twenty minutes.

Another method is soaking equipment for twenty minutes in a solution made of one part chlorine bleach to seven parts water, or in seventy percent ethanol alcohol. These solutions should be prepared fresh daily because they lose strength over time. When sterilizing a syringe with solution, pull some of the liquid inside and then push it out to clean the inside properly.

When caring for someone with an infectious disease, wash your hands frequently with soap and water.

First Aid

Basic Cleanliness and Protection

To prevent the spread of germs, always wash your hands with soap and water before and after giving first aid. When someone is injured, helping them is the first priority, but you must also protect yourself from HIV and other blood-borne diseases.

If a person is bleeding, first encourage them to press firmly on their own wound to stop the bleeding. If they cannot do this, protect your hands by wearing gloves or covering your hands with a clean plastic bag. Place a clean, thick cloth directly over the wound and apply firm pressure.

Avoid touching objects covered with blood. Be careful not to injure yourself with needles or sharp objects. Cover any cuts or wounds on your own body with clean, dry bandages. Extra care is needed when helping many injured people, such as after an accident or fight.

If blood or body fluids touch your skin, wash immediately with soap and water. If fluids enter your eyes or other sensitive areas, rinse thoroughly with plenty of clean water.

Fever

Fever means the body temperature is higher than normal. It is not a disease itself but a sign of illness. A high fever above 39°C can be dangerous, especially for small children.

When a person has a fever, remove most or all clothing. Small children should be left undressed until the fever decreases. Fresh air and a light breeze help lower body temperature and do not cause harm.

Give medicine to reduce fever. Aspirin can be used, but for children it is safer to give acetaminophen (paracetamol). Always be careful with the correct dose.

A person with fever should drink plenty of liquids such as water or juice. For babies and small children, water should be boiled and cooled first. Make sure the child urinates regularly. Dark urine or little urine means more fluids are needed.

If possible, find and treat the cause of the fever.

Very High Fever

A very high fever above 40°C must be reduced quickly because it can cause seizures or brain damage, especially in children.

Place the person in a cool area and remove all clothing. Fan the person. Pour cool, not cold, water over the body or place cloths soaked in cool water on the chest and forehead. Continue until the temperature drops below 38°C.

Give cool liquids to drink. Give aspirin or acetaminophen according to age and dosage instructions.

If the person cannot swallow tablets, crush them, mix with water, and give rectally using a syringe without a needle.

Shock

Shock is a life-threatening condition that can result from heavy bleeding, severe burns, dehydration, serious illness, or severe allergic reaction. Internal bleeding can also cause shock.

Signs of shock include a weak and rapid pulse, pale cold sweaty skin, very low blood pressure, confusion, weakness, or unconsciousness.

At the first sign of shock, loosen tight clothing. Lay the person down with their feet slightly higher than their head. If there is a serious head injury, keep the person in a half-sitting position instead.

Stop any bleeding while protecting your hands. Keep the person warm with a blanket if they feel cold. If conscious and able to drink, give small sips of water or rehydration fluids. If dehydration is severe and you are trained, give intravenous fluids.

Treat any wounds. For pain, give aspirin or another pain reliever but avoid medicines that cause sleepiness, such as codeine.

Stay calm, reassure the person, and seek medical help quickly.

If the person is unconscious, lay them on their side with the head low to prevent choking. If vomiting occurs, clear the mouth immediately and keep the head tilted to one side so vomit does not enter the lungs. If a neck or spine injury is suspected, avoid moving the head or back.

Do not give anything by mouth to an unconscious person. If trained, give intravenous saline solution rapidly and seek urgent medical care.

Loss of Consciousness

Loss of consciousness can happen for many reasons. Common causes include drunkenness, a strong hit to the head, fainting due to fear or weakness, low blood sugar, shock, heat stroke, seizures, stroke, poisoning, and heart attack.

If a person is unconscious and the cause is unknown, check immediately whether the person is breathing properly. If breathing is weak or absent, tilt the head back and pull the jaw and tongue forward to open the airway. Remove anything stuck in the throat. If the person is not breathing, begin mouth-to-mouth breathing immediately.

Check for heavy bleeding and control it if present. Look for signs of shock such as pale, moist skin and a weak, rapid pulse. If shock is suspected, lay the person down with the head lower than the feet and loosen tight clothing.

Consider heat stroke if the skin is hot, red, dry, and the person has a very high fever without sweating. In that case, move the person into shade, keep the head slightly raised, soak the body with cold water, and fan continuously.

Positioning an unconscious person depends on the condition. If the skin is very pale, as in shock or fainting, keep the head lower than the feet. If the skin is red or normal, as in heat stroke, stroke, or head injury, keep the head slightly raised.

If there is any chance of serious injury, especially to the neck or back, avoid moving the person unless absolutely necessary. Move carefully and do not bend the spine. Look for wounds or broken bones while keeping movement minimal.

Choking

If something gets stuck in a person’s throat and they cannot breathe, stand behind them and wrap your arms around their waist. Place your fist above the navel and below the ribs and give a strong upward thrust. This pushes air from the lungs and may remove the blockage. Repeat if necessary.

If the person is larger than you or already unconscious, lay them on their back. Place your hands between the navel and ribs and give strong upward pushes. For pregnant women, overweight persons, small children, or wheelchair users, press on the chest instead of the abdomen. If breathing does not return, begin mouth-to-mouth breathing.

Drowning

A person who has stopped breathing has only about four minutes to survive without brain damage. Begin mouth-to-mouth breathing immediately, even before removing the person fully from the water if it is shallow enough to stand safely.

If air does not enter the lungs, once on shore place the person on their side with the head lower than the feet and press on the abdomen to remove water, then immediately continue mouth-to-mouth breathing.

Mouth-to-Mouth Breathing

Breathing can stop due to choking, blocked airway from the tongue or mucus, drowning, smoke inhalation, poisoning, strong injury to the head or chest, or heart attack.

First, remove any visible object from the mouth or throat and pull the tongue forward. Lay the person on their back, tilt the head back gently, and pull the jaw forward to open the airway.

Pinch the nostrils closed, seal your mouth over theirs, and blow air into the lungs until the chest rises. Allow the air to come out, then repeat about every five seconds. For babies and small children, cover both nose and mouth with your mouth and breathe gently about every three seconds.

Continue until the person breathes independently or until there is no doubt that life cannot be restored. Mouth-to-mouth breathing does not transmit HIV unless there is active bleeding in the mouth.

Emergencies Caused by Heat

Heat cramps occur in hot weather when a person sweats heavily and loses salt. Painful cramps may appear in the legs, arms, or stomach. Treatment involves resting in a cool place and drinking water mixed with salt. Gentle massage can help relieve pain.

Heat exhaustion happens when a person becomes weak and pale from working in hot conditions. The skin is cool and moist, and the pulse is rapid and weak. Body temperature is usually normal. Treatment includes lying down in a cool place, raising the feet, and drinking salt water if conscious.

Heat stroke is rare but extremely dangerous. It is more common in older adults, overweight individuals, and alcoholics during very hot weather. The skin becomes red, hot, and dry. The body temperature can rise above 42°C, and the person may lose consciousness. Immediate cooling is critical. Move the person to shade, soak with cold or ice water, fan continuously, and seek urgent medical care.

How to Control Bleeding from a Wound

To control bleeding, first raise the injured part above the level of the heart if possible. Place a clean, thick cloth directly over the wound and press firmly. If no cloth is available, use your hand. Keep steady pressure without stopping to check the wound. Bleeding may take twenty minutes or even longer to stop. Direct pressure alone can stop most bleeding, even in very serious injuries.

If direct pressure does not stop the bleeding, continue pressing firmly and keep the injured part raised. Apply a tight bandage or clean clothing to maintain pressure. You may also press on pressure points where an artery can be pushed against a bone to reduce blood flow. Maintain pressure for at least twenty minutes before checking if bleeding has stopped.

Do not use a tourniquet, rope, or wire to stop bleeding, as this can cause permanent damage and loss of the limb. Never apply dirt, kerosene, lime, coffee, or other substances to a wound. If bleeding is severe, raise the feet and lower the head to help prevent shock. Protect yourself from contact with blood, especially if you have cuts or sores on your hands.

How to Stop Nosebleeds

When someone has a nosebleed, have them sit upright and remain calm. Gently blow the nose to clear mucus and clots. Then pinch the soft part of the nose firmly for at least ten minutes without releasing pressure. The head should remain upright or slightly forward, not tilted back.

If bleeding continues, gently pack the nostril with clean cotton, leaving part of it outside for removal. Wetting the cotton with petroleum jelly or a suitable medicine may help. Pinch the nose again for ten minutes or longer. Leave the cotton in place for several hours after bleeding stops, then remove it carefully.

In some older individuals, bleeding may come from the back of the nose and may not stop with pinching. In this case, the person should lean forward and avoid swallowing until bleeding stops.

To prevent frequent nosebleeds, apply a small amount of petroleum jelly inside the nostrils twice daily. Sniffing lightly salted water may also help. Eating fruits such as oranges and tomatoes can help strengthen blood vessels.

Cuts, Scrapes, and Small Wounds

Before treating a wound, wash your hands thoroughly with soap and water. If the wound is bleeding, wear gloves or use plastic bags to protect your hands. Wash the skin around the wound with soap and cool boiled water.

Clean the wound itself with cool boiled water. If it contains dirt, you may use soap carefully, but rinse well because soap can irritate tissue. Remove all dirt from the wound, including under any loose skin. Use clean tweezers or sterile cloth if needed, and boil them first to ensure they are sterile. If available, use a syringe to gently flush the wound with clean boiled water.

Any dirt left inside the wound can cause infection. After cleaning, apply a thin layer of antibiotic cream if available. Cover the wound with clean gauze or cloth, allowing some air circulation. Change the dressing daily and check for signs of infection such as redness, swelling, warmth, pus, or increasing pain.

If the wound is dirty or caused by a puncture and the person has not had a tetanus immunization, they should receive one within two days.

Large Cuts and Wound Closure

A clean, recent cut heals faster when the edges are brought together. However, close a deep cut only if it is less than twelve hours old, very clean, and no medical professional is available to treat it the same day.

Before closing the cut, wash it thoroughly with cool boiled water and ensure no dirt or soap remains. If possible, flush it with a syringe.

One simple method to close a cut is using adhesive tape in a butterfly pattern to hold the edges together gently without sealing in dirt.

Stitches or Sutures with Thread

Before stitching a wound, check whether the edges of the skin come together naturally. If they close by themselves, stitches are usually not needed.

To stitch a wound, first boil a sewing needle and thin thread, preferably nylon or silk, for twenty minutes to sterilize them. Wash the wound thoroughly with cool, boiled water. Wash your hands carefully with soap and boiled water.

Begin stitching by placing the first stitch in the middle of the cut and tying it firmly. If the skin is tough, hold the needle with boiled pliers or a needle holder. Continue placing stitches along the cut until the edges are completely closed. Leave stitches in place for five to fourteen days depending on the location. On the face, remove stitches after five days. On the body, remove them after ten days. On the hands or feet, remove them after fourteen days. To remove a stitch, cut one side of the thread near the knot and gently pull it out.

Only close wounds that are very clean and less than twelve hours old. Do not close old, dirty, infected wounds, or bites from humans or animals. Closing these wounds can trap infection inside and cause serious complications. If a stitched wound shows signs of infection, remove the stitches immediately and leave the wound open.

Bandages

Bandages help keep wounds clean. Always use clean cloth or sterile gauze. Cloth used for bandages should be washed and dried in a clean place, preferably in sunlight or ironed. Before applying a bandage, clean the wound properly.

If sterile gauze is not available, you can prepare your own by wrapping clean cloth in thick paper, sealing it, and baking it in an oven for twenty minutes. Placing water in the oven can prevent burning.

If a bandage becomes wet or dirty, remove it, clean the wound again, and apply a fresh bandage. Change the bandage daily. Do not wrap a bandage too tightly around a limb because it can stop blood circulation. Many small cuts and scrapes heal best when cleaned and left open to air.

Infected Wounds

A wound is infected if it becomes red, swollen, warm, painful, produces pus, or has a bad smell. Infection may be spreading if the person develops fever, a red line appears above the wound, or nearby lymph nodes become swollen and tender.

Swollen lymph nodes in different areas may indicate infection in nearby body parts. For example, swollen nodes behind the ear may suggest scalp infection. Nodes in the neck may indicate infection of the ear, face, or throat. Nodes in the armpit may indicate infection in the arm or breast. Nodes in the groin may indicate infection in the leg, foot, or genitals.

To treat an infected wound, apply hot compresses for twenty minutes four times daily. Keep the affected area raised and at rest. In severe cases, give antibiotics such as penicillin and metronidazole, especially if tetanus vaccination is not up to date.

If the wound smells very bad, releases brown or gray fluid, or the skin turns black with bubbles or blisters, this may be gangrene and requires urgent medical attention.

High Risk Wounds

Some wounds are more likely to become dangerously infected. These include dirty wounds, puncture wounds, deep wounds that do not bleed much, wounds caused in areas where animals are kept, crushed or badly bruised wounds, bites from animals or humans, and bullet wounds.

These wounds require special care. Clean them thoroughly with boiled water and soap. Remove dirt, blood clots, and dead tissue. If the wound is deep, caused by a bite, or may still contain dirt, give antibiotics such as ceftriaxone or another cephalosporin for three to seven days. If these are not available, alternatives include erythromycin, sulfamethoxazole with trimethoprim, or other sulfa drugs.

Do not close high risk wounds with stitches or adhesive bandages. Leave them open. A skilled health worker may close them later if safe.

Tetanus risk is high in unvaccinated persons. After a high risk wound, give penicillin or ampicillin immediately. In severe wounds, larger doses may be needed for a week or more. Tetanus antitoxin may also be required. If the wound is from an animal bite and rabies is possible, seek immunization immediately.

Bullet, Knife, and Other Serious Wounds

Deep bullet or knife wounds carry a high risk of infection. Start antibiotics such as penicillin or ampicillin immediately. If the person has not been vaccinated against tetanus, give tetanus antitoxin and vaccination if possible. Seek medical help.

For bullet wounds in arms or legs, control heavy bleeding first. If bleeding is mild, allowing brief bleeding may help clean the wound. Wash only the outside surface with boiled water and avoid inserting objects into the wound. Apply a clean bandage and give antibiotics.

If there is a chance that a bone is broken, do not allow weight on the limb. Splint the limb and keep it still for several weeks.

In serious wounds, keep the injured person still and raise the injured part slightly above heart level.

Deep chest wounds are dangerous and require urgent medical help. If air is entering through the wound during breathing, cover it immediately with gauze coated in petroleum jelly or fat and wrap it tightly. If breathing worsens, loosen the bandage. Place the person in a comfortable position. Treat shock if present and give antibiotics and pain relief.

For bullet wounds to the head, place the person in a half sitting position, cover the wound with a clean bandage, give antibiotics, and seek medical help immediately.

Deep Wounds in the Abdomen

Any wound that enters the abdomen is very dangerous and requires immediate medical attention. While arranging transport to a hospital, cover the wound with a clean bandage.

If part of the intestines is protruding from the wound, do not try to push them back inside. Instead, cover them with a clean cloth soaked in lightly salted, cool, boiled water. Keep the cloth moist at all times.

Do not give the injured person anything by mouth, including food or water, unless reaching medical care will take more than two days. In that rare case, give only small sips of water. If the person is conscious and very thirsty, allow them to suck on a cloth soaked in water.

If the person shows signs of shock, raise the feet higher than the head. Give antibiotics by injection if available. Never give an enema, even if the abdomen becomes swollen or there is no bowel movement. If the intestine is torn, an enema can be fatal.

Do not delay. The person needs urgent transport to a hospital for surgery.

Emergency Problems of the Gut (Acute Abdomen)

Acute abdomen refers to sudden, severe abdominal conditions that often require urgent surgery to prevent death. Examples include appendicitis, peritonitis, and intestinal obstruction. In women, pelvic inflammatory disease or an ectopic pregnancy can also cause similar symptoms. The exact cause may not be clear until surgery is performed.

Signs suggesting a serious acute abdomen include continuous severe pain that worsens, constipation with vomiting, a swollen and hard abdomen, and a very ill appearance. Less serious illnesses may cause cramping pain that comes and goes, diarrhea, mild infection symptoms, previous similar episodes, and moderate illness.

Obstructed Gut

An obstructed intestine occurs when something blocks the passage of food and stool. Causes may include a mass of roundworms, a loop of intestine trapped in a hernia, or one part of the intestine sliding into another part. Any suspected obstruction is life-threatening and requires immediate hospital care.

Appendicitis and Peritonitis

Appendicitis is infection of the appendix located in the lower right abdomen. If the infected appendix bursts, it can cause peritonitis, which is a severe infection of the lining of the abdominal cavity.

The main sign of appendicitis is steady abdominal pain that gradually worsens. The pain often begins near the navel and later moves to the lower right side. Other symptoms may include loss of appetite, vomiting, constipation, and mild fever.

To check for appendicitis or peritonitis, ask the person to cough and observe whether this causes sharp abdominal pain. Another test involves pressing slowly on the abdomen and then quickly releasing. If sharp pain occurs when the hand is removed, this rebound pain suggests appendicitis or peritonitis.

If appendicitis or peritonitis is suspected, seek immediate medical care. Do not give food, drink, or an enema. Only small sips of water or oral rehydration solution may be given if dehydration begins. The person should rest quietly in a half sitting position. In advanced peritonitis, the abdomen becomes hard and extremely painful to touch. This is a life-threatening emergency.

Burns

Most burns can be prevented by keeping children away from fires, hot objects, and matches. Turn pot handles inward on stoves to prevent accidents.

Minor burns without blisters should be placed immediately in cold water to reduce pain and damage. Pain relief such as acetaminophen may be used, but aspirin should not be given to children.

Burns that cause blisters should not have the blisters broken. Do not apply ice. If blisters break, gently wash the area with soap and cooled boiled water. Sterilized petroleum jelly may be applied to sterile gauze and placed loosely over the burn. If petroleum jelly is unavailable, leave the burn uncovered. Never apply grease or butter.

If infection develops, indicated by pus, bad smell, fever, or swollen lymph nodes, apply warm salt water compresses three times daily. The water and cloth must be boiled before use. Dead tissue should be removed carefully. Antibiotic ointment may be applied, and oral antibiotics considered in severe cases.

Deep burns that destroy skin and expose raw or charred tissue are always serious. Burns covering large areas are also dangerous. Seek immediate medical care. Until help is available, cover the burn with a very clean cloth moistened with clean water. If medical care is not available, keep the burn clean, loosely covered, and change dressings frequently. Antibiotics such as penicillin may be given. Never apply grease, fat, herbs, or other harmful substances. Honey can be applied to help prevent infection and should be replaced at least twice daily after gentle cleaning.

Special Precautions for Very Serious Burns

A person with severe burns can easily go into shock because of pain, fear, and loss of body fluids from the wound. It is important to comfort and reassure the person. Give pain relief such as acetaminophen or aspirin, but avoid aspirin in children. Codeine may be given if available. Bathing open burns in slightly salted cool, boiled water can also help reduce pain. Use one teaspoon of salt per liter of water.

The burned person must drink plenty of fluids. If the burned area is large, prepare a drink by adding half a teaspoon of salt and half a teaspoon of baking soda to one liter of water. Add two or three tablespoons of sugar or honey, and some orange or lemon juice if available. The person should drink this solution frequently until urination becomes regular. For large burns, about four liters per day may be needed. For very large burns, up to twelve liters per day may be necessary.

People with severe burns should eat protein rich foods to help healing. No specific foods need to be avoided.

Burns Around the Joints

Burns near joints such as between fingers or in the armpit require special care. Place sterile gauze pads with petroleum jelly between burned surfaces to prevent them from sticking together while healing. Joints should be straightened completely several times a day to prevent stiff scars that limit movement. This may be painful but is important. A healing burned hand should be kept in a slightly bent position.

Broken Bones (Fractures)

When a bone is broken, the most important step is to keep it still. This prevents further injury and helps healing. Before moving the injured person, use splints, bark, cardboard, or other firm materials to keep the bone from moving. A plaster cast can later be applied at a health center.

If the bones appear to be in correct position, do not attempt to move them. If the bones are badly out of position and the break is recent, you may attempt to gently straighten them before applying a cast. The sooner this is done, the easier it is. Muscle relaxing medicine such as diazepam or pain relief such as codeine may be given beforehand if available.

To set a broken wrist, apply slow and steady pulling force to the hand for several minutes to separate the bone ends. While one person maintains steady traction, another person gently aligns the bones. Avoid sudden movements or forceful jerks. Setting bones incorrectly can cause serious harm and should ideally be done by someone experienced.

Healing time depends on age and severity. Children heal quickly, while older people heal more slowly. A broken arm usually requires a cast for about one month, with limited use for another month. A broken leg may require about two months in a cast.

Broken Thigh or Hip

A broken thigh or hip requires careful handling. The whole body should be splinted to prevent movement, and the person must be taken to a health center immediately.

Broken Neck or Back

If there is any suspicion of neck or spine injury, do not change the person's position unless absolutely necessary. Avoid bending the neck or back. Seek trained medical help before moving the person whenever possible.

Broken Ribs

Broken ribs are very painful but usually heal without special treatment. Do not tightly bind the chest. Pain relief and rest are recommended. To prevent lung problems, take several deep breaths every two hours despite discomfort. If there is coughing of blood, difficulty breathing, or a rib breaks through the skin, give antibiotics and seek medical care.

Open Fractures

When a broken bone pierces the skin, the risk of infection is very high. Wear gloves or protective coverings on your hands. Clean the wound and exposed bone gently but thoroughly with cool, boiled water. Cover with a clean cloth. Do not push the bone back inside until it is completely clean. Splint the limb to prevent further injury. Give antibiotics such as penicillin, ampicillin, or tetracycline immediately to reduce infection risk.

How to Move a Badly Injured Person

When moving a seriously injured person, lift them very carefully without bending the body. Take special care to keep the head and neck straight. If there is a possible neck injury, place tightly folded clothing or sandbags on both sides of the head to prevent movement. While carrying the person, try to keep the feet slightly raised, even when going uphill.

Dislocations

A dislocation occurs when a bone comes out of place at a joint. Treatment has three main goals. First, try to return the bone to its normal position as soon as possible. Second, keep the joint firmly bandaged for about one month to prevent it from slipping out again. Third, avoid heavy use of the limb for two to three months so the joint can heal fully.

To set a dislocated shoulder, have the person lie face down on a firm surface with the affected arm hanging down. Apply steady downward pulling force on the arm for fifteen to twenty minutes, then gently release. The shoulder may move back into place. Another method is to attach a weight of ten to twenty pounds to the arm and allow steady traction for fifteen to twenty minutes.

After the shoulder returns to position, bandage the arm firmly against the body for one week to one month. To prevent stiffness, remove the bandage briefly three times daily and gently move the arm in small circles while it hangs at the side. Avoid lifting heavy objects for at least one month. If the joint cannot be returned to place, seek medical help immediately.

Strains and Sprains

It is often difficult to tell whether a joint is bruised, sprained, or broken without an X-ray. However, initial treatment is similar. Keep the joint still and provide firm support with a bandage or splint. If the foot is injured, use crutches to reduce weight bearing. Severe sprains usually require three to four weeks to heal.

To reduce pain and swelling, keep the injured area raised. During the first one or two days, apply cold compresses or ice wrapped in cloth for twenty to thirty minutes every hour. After twenty four to forty eight hours, once swelling stops increasing, begin warm soaks several times daily.

An elastic bandage can help support the joint and reduce swelling. Wrap from the toes upward if treating an ankle. Ensure the bandage is not too tight and remove it briefly every one or two hours. Pain relief such as acetaminophen may be used. Do not massage or rub a sprain or fracture, as this can cause harm. If pain and swelling do not improve within forty eight hours, or if the joint feels unstable, seek medical help.

Poisoning

Many poisonings occur in children who swallow harmful substances. All poisons should be kept out of reach of children. Never store dangerous liquids such as kerosene or gasoline in soft drink bottles, as children may mistake them for beverages.

Common poisons include insecticides, medicines taken in excess, iron tablets, iodine, bleach, cigarettes, alcohol, poisonous plants, matches, fuels, cleaning chemicals, and spoiled food.

If poisoning is suspected and the person is awake and alert, induce vomiting by stimulating the throat or giving water mixed with mild soap or salt. Activated charcoal mixed with water can help absorb poison. Adults may require a larger amount.

Do not induce vomiting if kerosene, gasoline, strong acids, corrosive chemicals, or lye have been swallowed, or if the person is unconscious. Instead, give water or milk to dilute the poison if the person is conscious. Keep the person warm but not overheated. Seek medical help immediately in severe cases.

Snakebite happens when a snake bites a person. It is important to find out whether the snake is poisonous or not. The bite of a poisonous snake usually leaves two deep fang marks. Sometimes there may also be small teeth marks. The bite of a non-poisonous snake usually leaves two rows of small teeth marks but no fang marks.

Many harmless snakes are often mistaken for poisonous ones. It is important to know which snakes in your area are truly dangerous. Boa constrictors and pythons are not poisonous. Non-poisonous snakes should not be killed because they help by killing mice and other harmful animals. Some even kill poisonous snakes.

If a person is bitten by a poisonous snake, the person should remain calm and avoid moving the bitten part. Movement allows the poison to spread faster through the body. If the bite is on the foot, the person should not walk. Medical help should be sent for immediately. Remove any jewelry because swelling may occur quickly.

Wrap the bitten area with a wide elastic bandage or clean cloth to slow the spread of poison. The bandage should be tight but not so tight that it stops blood flow. You should still be able to feel the pulse. If you cannot feel the pulse, loosen the bandage slightly. Wrap the bandage over the hand or foot and up the entire arm or leg. Then use a splint to keep the limb from moving. Carry the person on a stretcher if possible to the nearest health center.

If possible, take the snake to the health center so doctors can identify it and give the correct antivenom. Leave the bandage on until the antivenom injection is ready. If there is no antivenom available, remove the bandage. Tetanus protection should also be given.

Do not cut the bite area. Do not tie a tight rope around the limb. Do not put ice on the bite. Do not try to suck out the poison. Do not give alcohol. These actions can make the situation worse.

The bite of a beaded lizard is treated in the same way as a poisonous snakebite, but there is no effective antivenom. Wash the bite area well. Keep the bitten part below the level of the heart and avoid movement. Seek medical help immediately.

Scorpion stings are usually not dangerous for adults, but they can be dangerous for children under five years old, especially if the sting is on the head or body. Adults can take pain medicine and apply ice to reduce pain. Hot compresses may help if pain lasts for a long time. In children, medical help should be sought quickly. In some places, antitoxin is available and must be given within two hours. If the child stops breathing, give mouth-to-mouth breathing. Tetanus protection should also be given.

Most spider bites are painful but not dangerous. However, bites from certain spiders such as the black widow can cause severe illness. A black widow bite may cause painful muscle cramps and severe abdominal pain. Pain medicine should be given and medical help sought. Some special medicines may be required in serious cases. Tetanus protection should also be provided. An antivenom exists but may not be easily available.

Good food is necessary for growth, strength, and good health. Many common illnesses happen because people do not eat enough food or do not eat the right kinds of food. A person who is weak or sick because of poor eating is called malnourished. Malnutrition can cause many health problems.

In children, poor nutrition can cause slow growth and failure to gain weight. Children may be slow in walking, talking, or thinking. They may have thin arms and legs with a swollen belly. They often get infections that are more severe and last longer. They may feel tired, sad, and uninterested in playing.

Swelling of the feet, hands, and face can occur. Hair may become thin, dry, lose color, or fall out. Vision may become poor at night, and in severe cases blindness can occur.

In people of any age, malnutrition can cause weakness, tiredness, loss of appetite, anemia, sores at the corners of the mouth, a sore tongue, numbness or burning in the feet, and easy bruising. Other problems like diarrhea, frequent infections, dry skin, bleeding gums, headaches, heart palpitations, anxiety, and liver disease can also be related to poor nutrition.

During pregnancy, poor nutrition can cause anemia and weakness in the mother. It increases the risk of miscarriage, stillbirth, small babies, and babies born with disabilities.

Poor nutrition weakens the body's ability to fight disease. Children who are malnourished are more likely to develop severe diarrhea and die from it. Diseases such as measles and tuberculosis are more dangerous in malnourished people. Even minor illnesses like colds can become more serious.

Good nutrition helps prevent disease and helps sick people recover. Sick children must continue to eat. They should be fed small amounts many times a day. They should drink plenty of fluids. If they cannot eat solid foods, the food should be mashed or given as soft porridge.

Signs of severe malnutrition may appear during illness. Swelling of the hands, feet, and face, dark spots on the skin, and peeling sores are warning signs. During and after sickness, eating well is very important.

Malnutrition can happen by not eating enough food, by not eating the right types of food, or by eating too much of certain foods. Children are especially at risk because they need food for growth. Pregnant and breastfeeding women need extra food. Elderly people may eat less but still need proper nutrition. People with HIV need more food to fight infection.

A malnourished child is often thin, short, irritable, and sick more often. Measuring the upper arm is one way to check nutrition. After one year of age, if the middle upper arm measures less than 13.5 cm, the child is malnourished. If it measures less than 12.5 cm, the child is severely malnourished. Regular weighing also helps monitor growth.

To prevent malnutrition, people need enough energy, body-building foods, and protective foods. In many places, people eat one main food such as rice, maize, millet, wheat, cassava, potato, banana, or breadfruit. This main food provides energy but is not enough alone.

Helper foods must be added to the main food. Young children should be fed frequently, at least five times a day if they are very young or underweight. High-energy helper foods such as vegetable oils, nuts, seeds, sugar, or honey can be added to increase energy. Body-building foods such as beans, milk, eggs, fish, meat, and groundnuts help growth and repair. Protective foods such as orange and yellow fruits and vegetables and dark green leafy vegetables provide important vitamins and minerals.

Eating the usual main food and adding available helper foods from each group helps maintain good health. Feeding children enough food and feeding them often is usually more important than the specific types of food.

Malnutrition is most common and most serious in children, especially in poor communities. Children need a lot of nutritious food to grow well and stay healthy. There are different forms of malnutrition.

Mild malnutrition is very common and often not easy to notice. The child may grow more slowly and gain weight more slowly than a well-nourished child. He may look small and thin but not appear seriously sick. However, because he is poorly nourished, his body cannot fight infections well. He may get sick more often and take longer to recover. Diarrhea and colds are more frequent and may become more serious. Diseases like measles and tuberculosis are more dangerous for these children. Regular weighing or measuring the middle upper arm helps detect mild malnutrition early so it can be treated before it becomes severe.

Severe malnutrition often occurs in babies who stop breastfeeding early and do not receive enough high-energy food. It may begin during or after illnesses such as diarrhea. There are two main types.

Dry malnutrition, also called marasmus, happens when a child does not get enough of any kind of food. The child becomes extremely thin, small, and weak. The body looks wasted, with very little fat or muscle. The child appears like skin and bones. This child needs more food, especially energy-rich foods.

Wet malnutrition, also called kwashiorkor, causes swelling of the feet, hands, and face. This can happen when a child does not get enough energy or protein. Sometimes moldy or spoiled foods may contribute to the problem. The child needs frequent feeding with energy-rich foods and some protein-rich foods. Old or moldy foods should be avoided.

Other forms of malnutrition occur when certain vitamins or minerals are missing. Lack of vitamin A can cause night blindness. Lack of vitamin D can cause rickets. Not eating enough fruits and vegetables can cause skin problems, sores in the mouth, and bleeding gums. Lack of iron can cause anemia. Lack of iodine can cause goiter.

Poor nutrition is often linked to poverty. Some families do not have enough land or money to grow or buy nutritious food. In some places, land is used to grow crops for sale instead of food for families. Fair sharing of land and income can improve nutrition in the long term.

Communities can improve nutrition by working together. They can improve farming methods, grow different crops, store food properly, raise fish or chickens, and create family gardens. Cooperation helps increase food supply at low cost.

When money is limited, it should be spent wisely. Buying nutritious foods such as eggs, milk, or beans is better than spending money on alcohol, tobacco, sweets, or soft drinks. Parents should focus on foods that improve the health of their children.

Many people eat large amounts of starchy main foods but do not add enough helper foods to provide extra energy, protein, vitamins, and minerals. This often happens because helper foods, especially animal foods like meat and milk, are expensive.

Animal foods usually require more land and resources to produce. For poor families, it is often better to grow or buy plant foods such as beans, peas, lentils, and groundnuts along with a main food like maize or rice instead of spending money on expensive meat or fish.

However, when possible, eating some animal foods is beneficial because plant proteins may not contain all the proteins the body needs. Eating a variety of plant foods helps provide a better balance of nutrients. For example, eating beans together with maize gives better nutrition than eating either alone. Adding fruits and vegetables improves the diet further.

Breast milk is the best and most complete food for babies. It is low cost, protects against disease, and provides all necessary nutrients. Mothers can eat plant foods and convert them into high-quality breast milk.

Eggs are often an affordable and good source of animal protein. They can be mixed into baby food if breast milk is not available or given along with breastfeeding as the baby grows. Boiled and finely ground eggshells can provide calcium for pregnant women who have weak teeth or muscle cramps. Chickens are also a good source of protein, especially if families raise them at home.

Organ meats such as liver, heart, kidney, and blood are rich in protein, vitamins, and iron. They are often cheaper than other meats. Fish is also nutritious and sometimes less expensive than meat.

Beans, peas, and lentils are low-cost sources of protein. Sprouting them before cooking increases their vitamin content. For babies, well-cooked beans can be mashed or strained. Growing legumes also improves soil quality, helping future crops grow better.

Dark green leafy vegetables are rich in vitamin A and contain iron and some protein. Leaves of sweet potato, beans, pumpkins, and similar plants are especially nutritious. They can be dried, powdered, and added to baby food. Light green vegetables such as lettuce have less nutritional value.

Cassava leaves are more nutritious than cassava roots and contain more protein and vitamins. Eating the leaves together with the root increases nutritional value at no extra cost.

Soaking maize in lime before cooking increases its calcium content and helps the body use certain vitamins and proteins more effectively.

Whole grains such as whole wheat and moderately milled rice contain more nutrients than highly processed white grains. Eating grains together with beans or lentils improves protein quality.

Vegetables and grains should be cooked in small amounts of water and not overcooked to prevent loss of nutrients. Leftover cooking water should be consumed or used in soups.

Many wild fruits and berries contain vitamin C and natural sugars, providing extra vitamins and energy. Care must be taken to avoid poisonous varieties.

Cooking in iron pots or adding a clean piece of iron to cooking food can increase iron content and help prevent anemia. Adding tomatoes improves iron absorption. Soaking iron nails in lemon juice and drinking the juice can also provide iron.

In some areas, low-cost baby food products made from soybean, dried fish, skim milk, or similar ingredients are available. These can be mixed with porridge or cereal to improve nutritional value at low cost.

People often ask whether vitamins should be taken as pills, injections, syrups, or from food. A person who eats a balanced diet that includes vegetables and fruits usually gets all the vitamins needed from food. It is always better to eat nutritious foods than to depend on vitamin pills or injections.

Sometimes food is scarce, or a person may already be poorly nourished or seriously ill, such as with HIV. In these cases, eating as well as possible is very important, and vitamin supplements taken by mouth may help. Vitamins taken by mouth work as well as injections, cost less, and are safer. Vitamin injections should generally be avoided.

There are many beliefs about foods that should not be eaten during sickness. Some people believe in “hot” and “cold” foods and avoid certain foods during illness. While some traditional beliefs may have value, many food restrictions can harm health. Sick people need more nutritious food, not less. Nutritious foods such as energy-rich foods, fruits, vegetables, legumes, nuts, milk, meat, eggs, and fish help recovery.

Certain harmful substances should be avoided, especially during illness. Alcohol can damage the liver, stomach, heart, and nerves. Smoking can cause lung diseases and worsen conditions like tuberculosis and asthma. Too much greasy food or coffee can worsen stomach problems. Excess sugar can reduce appetite and damage teeth, although small amounts may provide needed energy. Too much salt can worsen high blood pressure and heart problems. Some conditions like diabetes and stomach ulcers require special diets.

For the first six months of life, a baby should receive only breast milk. Breast milk provides complete nutrition and protects against diarrhea and infections. Babies do not need extra water or teas, even in hot weather. A mother's milk remains nutritious even if she is thin or weak.

If a mother has HIV, there is a risk of passing HIV to the baby through breast milk. However, in areas without clean water, avoiding breastfeeding may increase the risk of death from diarrhea and malnutrition. Medicines are now available to reduce the risk of HIV transmission. Families must consider their local conditions when deciding.

Most mothers can produce enough breast milk if they breastfeed often, eat well, and drink enough fluids. Babies should not be given other foods before six months of age. If breast milk is insufficient, the baby should continue breastfeeding and be given additional milk by cup, not bottle. Boiled cow's milk, goat's milk, evaporated milk, or powdered milk may be used. Sugar or vegetable oil can be added for extra energy. Condensed milk should not be used.

Milk and water should be boiled and cooled before use. Cups are safer than bottles because bottles are harder to clean and can cause infections. If bottles are used, they must be boiled each time.

If milk is not available, porridge made from rice, cornmeal, or other cereals can be given. Protein foods such as beans, eggs, meat, or chicken should be added and mashed well. Sugar and oil can also be added to increase energy.

From six months to one year of age, breastfeeding should continue and may be continued until the child is two or three years old if possible. When the baby reaches six months, other foods should be introduced in addition to breast milk. The baby should always breastfeed first and then receive other foods.

The first foods can be simple gruel or porridge made from the main local food such as maize meal or rice cooked in water or milk. Cooking oil should be added for extra energy. After a few days, other nutritious foods can be added one at a time in small amounts to avoid digestion problems. These foods should be well cooked and mashed. At first, they can be mixed with breast milk to make swallowing easier.

Young children have small stomachs and cannot eat large amounts at once. Therefore, they should be fed small amounts frequently, at least five times a day, with snacks between meals. Adding oil increases the energy content so the child gets enough calories before feeling full. Dark green leafy vegetables and other iron-rich foods should also be added when possible.

The period from six months to two years is when children are most at risk of malnutrition. After six months, breast milk alone is not enough to meet energy needs. If breastfeeding stops early, the risk of malnutrition increases. During this age, children should continue breastfeeding, receive nutritious complementary foods, eat frequently, and drink clean, purified water. Food and surroundings must be kept clean. When sick, children should be fed more often and given extra fluids.

For mothers infected with HIV, after six months it may be safer to switch from breastfeeding to other milks and foods if conditions allow, because the risk of HIV transmission through breast milk continues.

After one year of age, children can eat the same foods as adults but should continue breastfeeding or drinking milk if possible. They should receive the main local food along with helper foods that provide energy, protein, vitamins, iron, and minerals. Serving the child in a separate dish helps ensure adequate intake.

Children should not be accustomed to eating too many sweets or soft drinks because these reduce appetite for nutritious foods and harm teeth. However, in situations where food is limited or bulky, adding small amounts of sugar and vegetable oil can increase energy intake.

Harmful beliefs about diet can damage health. Some communities restrict nutritious foods for women after childbirth. This weakens the mother and may lead to anemia, infection, or serious complications. After childbirth, mothers need main foods along with protein-rich foods such as beans, eggs, chicken, milk, meat, and fish, as well as fruits, vegetables, and energy-rich foods. These foods strengthen and protect the mother.

It is not true that fruits like oranges or guavas worsen colds or coughs. These fruits contain vitamin C, which may help fight infections. It is also not true that foods such as pork or spices cannot be eaten while taking medicines. However, people with stomach problems should avoid excessive greasy foods.

Anemia occurs when the blood becomes thin due to blood loss, destruction of red blood cells, or lack of iron in the diet. Causes include bleeding, malaria, hookworm infection, chronic diarrhea, and poor nutrition. Women are at risk during menstruation, pregnancy, and childbirth. Children may develop anemia if iron-rich foods are not introduced after six months.

Signs of anemia include pale skin, pale eyelids and gums, white fingernails, weakness, and fatigue. Severe anemia may cause swelling, rapid heartbeat, and shortness of breath. A craving for dirt is common in anemic individuals.

Treatment and prevention include eating iron-rich foods such as meat, liver, fish, chicken, beans, lentils, and dark green leafy vegetables. Cooking in iron pots may help increase iron intake. Eating fruits and raw vegetables with meals improves iron absorption, while tea and coffee reduce it. Moderate to severe anemia requires iron tablets taken by mouth. Injections are usually unnecessary and can be dangerous. Underlying diseases must also be treated. Severe cases require medical care. Women should allow sufficient time between pregnancies to regain strength.

Rickets is a condition in children caused by lack of vitamin D, often due to insufficient sunlight exposure. It can cause bowed legs and bone deformities. Prevention includes daily exposure to sunlight for at least ten minutes without causing burns. Fortified milk and vitamin D supplements may help, but large doses of vitamin D can be harmful.

Sunlight is the best and simplest prevention and treatment for rickets.

High blood pressure, also called hypertension, can lead to serious health problems such as heart disease, kidney disease, and stroke. People who are overweight are more likely to develop high blood pressure.

Dangerously high blood pressure may cause frequent headaches, pounding of the heart, shortness of breath with mild activity, weakness, dizziness, and sometimes pain in the left shoulder or chest.

However, these signs can also occur in other diseases. High blood pressure often causes no symptoms at first, so regular measurement with a blood pressure cuff is important, especially for overweight individuals.

Prevention and care include losing excess weight, reducing fatty and greasy foods, avoiding animal fats such as pig fat, and using vegetable oil instead. Food should be prepared with little or no salt. Smoking should be stopped, and alcohol, coffee, and tea should be limited. If blood pressure remains very high, medicines may be required. Relaxation and weight loss can significantly help reduce blood pressure.

Being excessively overweight is unhealthy. Very heavy people are at greater risk of high blood pressure, stroke, gallstones, diabetes, arthritis, and other health problems. Changes in diet, especially increased intake of processed and junk foods high in calories and low in nutrition, contribute to unhealthy weight gain.

Weight loss can improve many health problems and make daily activities easier. Healthy weight loss can be achieved by reducing greasy and fatty foods, limiting sugar and sweets, increasing physical activity, avoiding processed foods, and eating more fresh fruits and vegetables.

Constipation occurs when a person has hard stools and does not have a bowel movement for three or more days. It is commonly caused by low fiber intake or lack of exercise. Drinking more water and eating fruits, green vegetables, and whole grain foods helps relieve constipation. Natural fiber sources include whole grain bread, cassava, wheat bran, rye, carrots, turnips, raisins, nuts, pumpkin seeds, and sunflower seeds. Adding a small amount of vegetable oil to food may also help. Regular walking or exercise is important, especially for older people.

If a person has not had a bowel movement for four or more days and does not have severe stomach pain, a mild salt laxative such as milk of magnesia may be used occasionally. Laxatives should not be used frequently. Laxatives should not be given to babies or young children. Severe constipation in babies may be relieved with a small amount of cooking oil inserted gently into the rectum.

Diabetes is a condition in which there is too much sugar in the blood. It may begin in childhood or adulthood. Diabetes that begins in young people is usually more serious and often requires insulin. Adult diabetes is more common after age forty, especially in overweight individuals.

Early signs of diabetes include constant thirst, frequent urination, tiredness, constant hunger, and weight loss. Later signs may include itchy skin, blurred vision, loss of feeling in hands or feet, frequent infections, sores that heal slowly, and in severe cases, loss of consciousness. These signs can also occur in other diseases, so testing urine for sugar is necessary. Special test strips can detect sugar in urine.

Adult diabetes can often be controlled with proper diet. Overweight diabetics should lose weight gradually. People with diabetes should avoid sugar, sweets, and sweet-tasting foods. High fiber foods such as whole grain bread are important. They should also eat moderate amounts of starchy foods like beans, rice, and potatoes, along with protein-rich foods.

Good hygiene is essential to prevent infection. Teeth should be cleaned after meals, skin kept clean, and shoes worn to prevent foot injuries. For poor circulation in the feet, resting with the feet elevated can help.

Acid indigestion and heartburn are often caused by eating too much heavy or greasy food, or by drinking excessive alcohol or coffee. These habits increase the production of stomach acid, leading to discomfort or a burning feeling in the upper abdomen or chest. Heartburn pain may be mistaken for heart disease, but if the pain worsens when lying down, it is more likely due to acid reflux.

Frequent or persistent acid indigestion may be a warning sign of a stomach ulcer. An ulcer is a chronic sore in the digestive system, usually caused by bacteria. Excess stomach acid prevents the sore from healing. Ulcer pain is often dull but can sometimes be sharp. It may improve temporarily after eating or drinking water but usually worsens an hour or more after meals, when meals are missed, or after consuming alcohol, fatty, or spicy foods. Pain is often worse at night.

Severe ulcers may cause vomiting, sometimes with blood. Fresh blood may appear red, while digested blood may look like coffee grounds. Blood in the stool from an ulcer usually appears black and tar-like. Some ulcers are silent and painless, and the first sign may be vomiting blood or passing black stools. This is a medical emergency and requires immediate medical attention.

Prevention and treatment of acid indigestion, heartburn, and ulcers include eating small, frequent meals instead of large ones and avoiding foods or drinks that worsen symptoms. Common triggers include alcohol, coffee, spicy foods, carbonated drinks, and fatty foods. Drinking plenty of water before and after meals and between meals may help reduce symptoms. Avoiding tobacco is important because it increases stomach acid production.

Antacids containing magnesium and aluminum hydroxide are generally safe and effective. If symptoms do not improve, treatment for ulcers may require antibiotics such as amoxicillin or tetracycline combined with metronidazole to treat the bacterial cause. Medicines like cimetidine or ranitidine reduce stomach acid and promote healing. Aloe vera is traditionally used in some places to soothe ulcers, although medical treatment is essential.

Some antacids, such as sodium bicarbonate and calcium-containing products, may provide temporary relief but can increase acid production later. These should not be used for ulcers. Medicines such as aspirin, iron salts, and corticosteroids can worsen ulcers and should be avoided if possible. Stress and tension can increase stomach acid, so relaxation may help. Early treatment is important to prevent complications such as bleeding or infection of the abdominal cavity.

A goiter is a swelling in the neck caused by enlargement of the thyroid gland. The most common cause is lack of iodine in the diet. Iodine deficiency during pregnancy can cause serious problems in babies, including mental impairment or deafness. Goiter is common in mountainous regions where natural iodine levels in soil and water are low. Eating large amounts of certain foods like cassava may increase the risk in these areas.

The best prevention for goiter is the use of iodized salt. Regular use of iodized salt prevents most goiters and may reduce existing ones. If iodized salt is not available, a single drop of tincture of iodine in a glass of water daily may be used, but excessive amounts are poisonous and dangerous. Seafood and seaweed contain iodine and may help, but iodized salt is the safest and easiest method.

People living in areas where goiter is common should avoid regular salt and always use iodized salt. They should also limit intake of foods like cassava or cabbage if goiter is developing. If a person with a goiter experiences trembling, nervousness, and bulging eyes, this may indicate toxic goiter, which requires medical evaluation.

Many sicknesses can be prevented by taking simple precautions. Eating well, maintaining personal and public cleanliness, and vaccinating children can prevent most common diseases before they begin.

Cleanliness plays a major role in preventing infections of the stomach, skin, eyes, lungs, and other parts of the body. Personal hygiene and sanitation in the community are equally important. Many intestinal infections spread because of poor hygiene and sanitation. Germs and worm eggs are present in the feces of infected individuals and can spread to others through dirty hands, contaminated food, or unsafe water.

Diseases spread through the feces-to-mouth route include diarrhea, dysentery, intestinal worms, hepatitis, typhoid fever, cholera, and sometimes polio. Transmission can occur easily when a person does not wash hands after using the toilet and then handles food. Even though worm eggs are too small to see, they can contaminate food and infect others.

Animals such as pigs, dogs, and chickens can also spread intestinal diseases if they enter homes or contaminate areas where children play. Poor sanitation and contact with animal feces increase the risk of infection for the whole family.

Preventive measures include using latrines, preventing animals from entering living areas, keeping children away from contaminated ground, and washing hands thoroughly after contact with feces and before preparing food. If diarrhea and worm infections are common in a community, hygiene practices are usually inadequate. When children die from diarrhea, poor nutrition often contributes to the severity.

Basic personal hygiene practices include washing hands with soap in the morning, after bowel movements, and before eating. Bathing regularly, especially in hot weather or after sweating, helps prevent skin infections, itching, and rashes. Even sick persons and babies should be bathed daily.

In areas where hookworm is common, walking barefoot should be avoided because hookworm larvae can enter through the skin of the feet and cause severe anemia. Daily brushing of teeth is important, especially after eating sweets. If a toothbrush is unavailable, teeth can be cleaned with salt and baking soda.

Home cleanliness is also important. Animals should not be allowed inside the house or areas where children play. Dogs should not lick children or sleep on beds, as they can spread disease. Human or animal feces near the home should be cleaned immediately, and children should be taught to use latrines.

Bedding should be placed in sunlight regularly. If bedbugs are present, bedding and sleeping areas should be cleaned thoroughly, including using boiling water if necessary. Lice and fleas should be removed from all family members because they carry disease. Animals carrying fleas should not be allowed indoors.

Spitting on the floor should be avoided because it spreads disease. When coughing or sneezing, the mouth should be covered with a hand or cloth. Regular cleaning of floors, walls, and furniture reduces hiding places for insects and pests that may spread illness.

Cleanliness in eating and drinking is very important for preventing disease. Ideally, all water that does not come from a safe system should be boiled, filtered, or purified before drinking. This is especially important for small children, people with HIV, and during outbreaks of diarrhea, typhoid, hepatitis, or cholera. However, having enough water is often more important than having perfectly pure water, especially for poor families.

A low-cost way to purify water is by placing it in a clean, clear bottle or plastic bag and leaving it in direct sunlight for at least six hours. If the water is cloudy, it should remain in the sun for at least two days. Sunlight helps kill most germs.

Food should always be protected from flies and insects because they carry germs. Food scraps and dirty dishes should not be left exposed. Food should be covered or stored in screened containers.

Fruit that falls on the ground should be washed thoroughly before eating. Children should not eat food that has dropped on the ground unless it is cleaned. Meat and fish should always be well cooked. Raw or undercooked pork can carry serious diseases. After handling raw chicken, hands should be washed before touching other foods.

Food that smells bad or is old should not be eaten because it may cause poisoning. Swollen or leaking canned foods should be avoided. Leftover cooked foods should be reheated thoroughly before eating. Freshly prepared foods are safest, especially for children, elderly people, and those who are very sick.

People with infectious diseases such as tuberculosis, influenza, or colds should eat separately from others. Their utensils should be washed carefully before reuse.

Protecting children's health requires additional care. Sick children should sleep separately from healthy children to prevent spread of infection. Children with contagious illnesses such as measles or whooping cough should be kept away from babies and young children.

People with long-term cough or tuberculosis should cover their mouths when coughing and should not sleep in the same room as children. They should seek medical treatment. Children living with someone who has tuberculosis should receive the BCG vaccine.

Children should be bathed regularly, have clean clothes, and keep their fingernails short, as germs and worm eggs can collect under long nails. Infectious diseases in children should be treated promptly to prevent spreading. Good nutrition strengthens the immune system and helps children resist infections.

Public cleanliness, or sanitation, is equally important. Wells and water sources must be kept clean and protected from animals. No one should defecate or throw garbage near water sources. Rivers and streams should remain clean upstream from drinking areas.

Garbage that can be burned should be burned. Other waste should be buried in pits far from homes and water sources.

Latrines should be built to prevent animals from reaching human waste. A deep pit with a simple structure over it works well. Throwing lime, dirt, or ashes into the pit after use reduces smell and flies. Latrines should be located at least 20 meters away from homes and water sources. If no latrine is available, defecation should take place far from living and water areas.

Closed latrines are better than open ones because they reduce flies and odor. They have a slab with a hole and a cover. Cement slabs are stronger and more durable than wood. Proper construction includes digging a deep pit, placing a reinforced slab over it, and keeping it covered when not in use.

Ventilated improved pit latrines use a ventilation pipe covered with a fly screen. The structure is kept dark inside so flies are drawn to the light at the top of the pipe, where they are trapped and die. This system reduces odors and fly breeding.

Worms and other intestinal parasites are common in many parts of the world. These parasites live in the intestines and can cause illness, especially in children. Some worms are large enough to be seen in the stool, while others are too small to be noticed without laboratory testing.

The most common intestinal worms include roundworm (Ascaris), pinworm (threadworm), whipworm (Trichuris), hookworm, and tapeworm. Roundworms, pinworms, and tapeworm segments can often be seen in the stool. Hookworms and whipworms may be present in large numbers without being visible.

Many worm infections spread through poor hygiene. Eggs from infected feces enter another person's mouth through contaminated hands, food, water, or soil. Preventing infection depends mainly on cleanliness, proper use of latrines, washing hands, wearing shoes, and cooking food thoroughly.

ROUNDWORM (Ascaris)
Roundworms are large, pink or white worms that can grow 20 to 30 cm long. Infection happens when a person swallows eggs from contaminated food, water, or dirty hands. After entering the body, the larvae travel through the bloodstream to the lungs, may cause coughing, then are swallowed again and grow in the intestines.

A heavy infection can cause stomach pain, indigestion, weakness, swollen belly in children, coughing, or rarely intestinal blockage. Prevention includes good sanitation and handwashing. Mebendazole is commonly used for treatment. Piperazine also works, but thiabendazole should not be used because it may cause worms to move dangerously into the airway.

PINWORM (Threadworm, Enterobius)
Pinworms are small, thin, white worms about 1 cm long. They lay eggs around the anus, causing itching, especially at night. Scratching spreads eggs to fingers, food, and other people.

Pinworms are not usually dangerous but can disturb sleep. Prevention includes keeping fingernails short, washing hands often, bathing regularly, changing clothes frequently, and wearing tight underwear at night. Mebendazole is effective, and all family members should often be treated at the same time. Good hygiene alone may eliminate the worms within several weeks.

WHIPWORM (Trichuris)
Whipworms are 3 to 5 cm long and spread through feces-to-mouth transmission. They usually cause mild symptoms but may lead to diarrhea. In severe cases, especially in children, part of the rectum may protrude. Prevention is the same as for roundworms. Mebendazole is used if treatment is needed.

HOOKWORM
Hookworms are small red worms that enter the body through bare feet when walking on contaminated soil. The larvae travel through the bloodstream to the lungs, are coughed up and swallowed, and then attach to the intestinal wall.

Hookworms can cause severe anemia, weakness, pale skin, and poor growth in children. Children who are very pale or who eat dirt may have hookworms. Wearing shoes helps prevent infection. Treatment includes mebendazole, albendazole, or pyrantel. Anemia should also be treated with iron-rich foods or supplements.

TAPEWORM
Tapeworms grow several meters long inside the intestine. Small white segments may be seen in the stool. Infection occurs from eating undercooked pork, beef, or fish containing cysts.

Most intestinal tapeworms cause mild stomach discomfort. However, if eggs enter the person's mouth from poor hygiene, cysts can form in the brain, causing headaches, seizures, or even death. This condition is very serious.

Prevention includes cooking meat thoroughly, especially pork. People with tapeworm infection must follow strict hygiene to prevent spreading eggs. Treatment includes niclosamide or praziquantel, taken according to instructions.

WORM MEDICINES
Many older worm medicines contain piperazine, which only works against roundworms and pinworms and should not be given to babies. Mebendazole is safer and treats several types of worms. Albendazole and pyrantel are also effective but may be more expensive. Thiabendazole can cause serious side effects and is usually not recommended.

The most important prevention for all intestinal worms is good sanitation, use of latrines, washing hands with soap, protecting food from flies, wearing shoes, and proper cooking of meat.

TRICHINOSIS

Trichinosis is caused by worms that burrow from the intestines into muscles. People get infected by eating undercooked or raw pork or other meat.

Symptoms:

From a few hours to 5 days after eating infected meat: diarrhea, nausea, stomach discomfort.

Severe cases may include fever, chills, muscle pain, swelling around eyes or feet, small bruises, and bleeding in the whites of the eyes.

Severe symptoms may last 3–4 weeks.

Treatment:

Seek medical help immediately.

Albendazole or mebendazole may be used. Cortico-steroids can help but only under medical supervision.

Prevention:

Only eat well-cooked pork and meat.

Do not feed raw meat scraps to pigs.

AMEBAS (AMEBIC INFECTION)

Amebas are microscopic parasites found in feces of infected people. They contaminate water or food and infect others.

Symptoms:

Diarrhea that comes and goes, sometimes alternating with constipation.

Cramping in the belly, mucus in stools, sometimes with blood.

Severe cases: much blood, weakness, illness.

May cause liver abscesses: tenderness or pain in the right upper belly, sometimes extending to the chest. If pus drains into the lungs, brown liquid may be coughed up.

Treatment:

Medical evaluation and stool analysis if possible.

Metronidazole, sometimes with diloxanide furoate.

For liver abscess, treat as above, then take chloroquine for 10 days.

Prevention:

Use latrines, protect drinking water, maintain cleanliness, eat well, avoid fatigue and alcohol.

GIARDIA

Giardia is a microscopic parasite that infects the gut, causing diarrhea—chronic or intermittent.

Symptoms:

Yellow, foul-smelling, frothy diarrhea without blood or mucus.

Swollen belly with gas, mild cramps, frequent burping and farting. Sulfur taste in burps.

Usually no fever.

Treatment:

Severe cases: metronidazole.

Quinacrine (Atabrine) may be used but has more side effects.

Good nutrition helps mild infections resolve naturally.

BLOOD FLUKES (SCHISTOSOMIASIS / BILHARZIA)

Blood flukes are worms that live in the bloodstream. They infect different body parts depending on the species.

Symptoms:

Blood in urine (common in Africa/Middle East) or bloody diarrhea (Africa, South America, Asia).

Lower belly pain, pain during urination, low fever, weakness, itching.

In women: sores resembling sexually transmitted infections.

Long-term infection can damage kidneys, liver, or spleen.

Sometimes no early signs.

Transmission:

Blood flukes require a water snail to complete their life cycle.

Infected people urinate or defecate in water → eggs hatch in snails → young worms leave snail → infect new people who swim, bathe, or wash in water.

Treatment:

See a health worker.

Praziquantel treats all types of blood flukes.

Metrifonate and oxamniquine work for certain types.

Prevention:

Use latrines and avoid urinating/defecating in or near water.

Support programs to eliminate snails and treat infected individuals.

Vaccinations, also called immunizations, protect children and adults from many dangerous diseases. Each country has its own vaccination schedule, and vaccines are usually given free of cost. It is much better to take children to a health center for vaccination while they are healthy than to wait until they become sick. Important vaccines include DPT, which protects against diphtheria, whooping cough, and tetanus. It is usually given as four or five injections starting at two months of age and continuing at four months, six months, eighteen months, and sometimes again between four and six years.

The polio vaccine protects against infantile paralysis. It is usually given as drops in the mouth four or five times. In some places the first dose is given at birth and the others are given at the same time as DPT. In other places, three doses are given with DPT, a fourth between twelve and eighteen months, and a fifth at four years of age. In families where someone has HIV, polio vaccine should be given by injection instead of drops.

The BCG vaccine protects against tuberculosis. It is given as a single injection under the skin of the left arm, either at birth or later. If someone in the household has tuberculosis, babies should be vaccinated in the first weeks or months of life. This vaccine usually causes a sore and leaves a scar.

The measles vaccine is given as one injection at nine months of age or older, and often a second dose at fifteen months or later. In many countries, a combined vaccine called MMR protects against measles, mumps, and rubella. It is given between twelve and fifteen months, with a second dose between four and six years. Measles vaccine should not be given to a child with HIV.

The hepatitis B vaccine is given in three injections, often at the same time as DPT. In some places the first dose is given at birth, the second at two months, and the third at six months. There should be at least four weeks between the first and second doses and eight weeks between the second and third.

The Hib vaccine protects against Haemophilus influenzae type b, which can cause meningitis and pneumonia in young children. It is usually given in three injections together with the first three DPT doses.

Tetanus toxoid vaccine, also called Td or TT, protects older children and adults against tetanus. One injection is recommended every ten years. In some places a dose is given between nine and eleven years of age and then repeated every ten years. Pregnant women should receive tetanus vaccination during each pregnancy to protect their newborn babies from tetanus.

The rotavirus vaccine is given by mouth two or three times, depending on the manufacturer, starting at two months of age. It protects against severe diarrhea, which is a leading cause of death in young children.

Some vaccines, such as those for measles, polio, and tuberculosis, must be kept frozen or very cold below eight degrees Celsius. Vaccines for hepatitis B, tetanus, and DPT must also be kept below eight degrees Celsius but should never be frozen. Prepared vaccines that are not used should be thrown away. DPT remains usable if it stays cloudy after preparation, but if it becomes clear or develops white particles, it is spoiled.

Preventing sickness and injury involves more than vaccination. Good hygiene, sanitation, proper nutrition, and wise use of medicines are also important. Communities can work together to improve conditions that cause poor health. Early and sensible treatment of illness is one of the best ways to prevent serious disease and death.

Certain habits can harm both individuals and those around them. Drinking alcohol in small amounts may not cause harm, but heavy drinking often leads to serious health and social problems. It can cause liver disease such as cirrhosis and hepatitis. Alcohol misuse can lead to poor judgment, family conflict, violence, and financial hardship. When money is spent on alcohol instead of food or living conditions, families suffer. People who recognize that alcohol is harming them or their families should admit the problem and seek support. Family, friends, and support groups can help. Communities can also organize efforts to reduce alcohol misuse and support those trying to stop drinking.

Smoking is dangerous for both the smoker and family members. It increases the risk of cancers of the lungs, mouth, throat, and lips. It causes chronic lung diseases such as bronchitis and emphysema and worsens asthma. Smoking can cause or worsen stomach ulcers and increases the risk of heart disease and stroke. Children of parents who smoke have more respiratory infections such as pneumonia. Babies born to mothers who smoked during pregnancy are often smaller and develop more slowly. Smoking also sets a harmful example for children and encourages them to start smoking. In addition, tobacco costs money that could be spent on food or other needs.

Carbonated drinks such as soft drinks and colas have little nutritional value except for sugar. They are expensive for the amount of nutrition they provide. When families spend money on these drinks instead of nutritious foods like eggs or fruits, children may become undernourished. Drinking many sweet beverages can also cause dental cavities and tooth decay at an early age. These drinks can worsen acid indigestion and stomach ulcers. Natural drinks made from fruits are healthier and usually less expensive than carbonated drinks.

Dehydration happens when the body loses more liquid than it takes in. It is most common during severe diarrhea, especially when there is vomiting. It can also happen in very serious illness when a person is too sick to drink or eat enough. Although people of any age can become dehydrated, small children become dehydrated more quickly and are at greater risk of death. Most children who die from diarrhea die because of dehydration.

Early signs of dehydration include thirst and passing very little dark yellow urine. Other signs are sudden weight loss, dry mouth, sunken eyes without tears, and in babies, a sunken soft spot on the head. The skin loses its elasticity and does not return quickly to normal when pinched. In very severe cases, there may be a rapid weak pulse, fast deep breathing, fever, or seizures. When a person has watery diarrhea, especially with vomiting, action should be taken immediately without waiting for these signs to appear.

To prevent or treat dehydration, give plenty of liquids. A rehydration drink is best, but thin cereal porridge, soups, teas, or clean water can also be used. Fluids should be given in small sips every few minutes, day and night, even if the person vomits, because some fluid will stay in the body. Breastfed babies should continue to receive breast milk frequently and before other drinks. Food should be given as soon as the person is willing to eat.

A home rehydration drink can be prepared using clean water mixed with small amounts of salt and sugar, or salt and powdered cereal that is boiled into a thin liquid. The drink should not taste saltier than tears. It should be freshly prepared and checked each time before giving, especially cereal drinks which can spoil quickly in hot weather. Large adults may need three or more liters a day, and small children usually need at least one liter a day or one glass for each watery stool. If dehydration worsens or danger signs appear, medical help is needed and intravenous fluids may be required.

Diarrhea means passing loose or watery stools. If there is mucus and blood in the stool, it is called dysentery. Diarrhea can be mild or severe, sudden or long-lasting. It is more common and more dangerous in poorly nourished children. Poor nutrition weakens the child, making diarrhea more frequent and severe. Diarrhea itself also worsens malnutrition, creating a harmful cycle in which each condition makes the other worse.

Common causes of diarrhea include infections from viruses, bacteria, parasites, and worms. It may also result from unsafe water, poor sanitation, food poisoning, certain medicines, difficulty digesting milk, allergies, infections in other parts of the body, malaria, or HIV.

Prevention of diarrhea depends on good nutrition and cleanliness. Safe water, use of latrines, protection of food from flies and dirt, and proper hygiene are important. Breastfeeding for the first six months helps protect babies from diarrhea. If breastfeeding is not possible, feeding with a cup and spoon is safer than using a bottle. New foods should be introduced gradually and mashed well. Babies should be kept clean and prevented from putting dirty objects in their mouths.

Most cases of diarrhea do not require medicine. The main treatment is preventing dehydration and meeting nutritional needs. Fluids must be given frequently, especially rehydration drink in severe cases. Food should be continued as soon as possible. Small frequent meals are better than large meals. Energy foods such as rice, bananas, potatoes, and cooked grains are helpful. Protein foods like eggs, chicken, fish, beans, and lentils should also be given. Breast milk should always be continued.

Breast milk is the best food during diarrhea. Other milks can usually be continued, although in rare cases they may worsen diarrhea. If that happens, smaller amounts can be given mixed with other foods while ensuring enough energy and protein intake.

Most anti-diarrhea medicines are not recommended. Medicines that stop diarrhea do not correct dehydration and may keep harmful germs inside the body. Some can cause harm, especially in small children. Laxatives and purges should never be used. Antibiotics are useful only in certain cases, such as some bacterial infections or amebic dysentery. If diarrhea contains mucus and blood and suggests amebic infection, specific treatment such as metronidazole may be needed under medical guidance.

Severe diarrhea with blood and fever is often caused by bacterial dysentery due to Shigella. In adults, ciprofloxacin can be given as a single oral dose of 1 gram. However, ciprofloxacin should not be used in pregnant women or in children under 18 years of age. For young infants, medical help should be sought. Shigella may be resistant to ampicillin or cotrimoxazole, although these medicines are sometimes still used. If there is no improvement within two days, another medicine should be tried or medical help should be sought. Cotrimoxazole should not be used during the last three months of pregnancy. Azithromycin is another effective option and is safe during pregnancy and in children. In adults, 500 mg is given on the first day, followed by 250 mg once daily for four days.

Severe diarrhea with fever but usually without blood may be partly due to dehydration. Large amounts of rehydration drink should be given. If the person does not improve within six hours after starting rehydration, medical help is needed. Typhoid fever should be considered and treated if signs are present. In areas where falciparum malaria is common, persons with diarrhea and fever should also be treated for malaria, especially if the spleen is enlarged.

Yellow, foul-smelling diarrhea with bubbles or froth and no blood or mucus may be caused by giardia infection or malnutrition. There may be excessive gas and bad-smelling burps. Treatment includes plenty of liquids, nutritious food, and rest. Severe giardia infections can be treated with metronidazole. Quinacrine is a cheaper alternative but has more side effects.

Chronic diarrhea that lasts a long time or keeps returning may be due to malnutrition or chronic infections such as amebiasis or giardiasis. The child should receive more nutritious food more frequently. If the condition does not improve, medical advice should be sought.

Very large amounts of watery stool that look like rice water may indicate cholera. Cholera often occurs in epidemics and can cause rapid severe dehydration, especially if vomiting is present. Continuous rehydration is essential. In severe cases, antibiotics such as doxycycline or tetracycline may be given, and erythromycin can be used for children. Cholera cases should be reported to health authorities, and medical help should be sought immediately.

Diarrhea is particularly dangerous in babies. Breastfeeding should continue, along with small frequent sips of rehydration drink. If vomiting occurs, give small amounts frequently. If there is no breast milk, diluted milk or milk substitutes can be given. If milk worsens diarrhea, other protein sources such as mashed chicken, eggs, lean meat, or skinned mashed beans mixed with carbohydrates and boiled water can be used. Babies younger than one month should be seen by a health worker before giving medicine. If no help is available and the baby is very sick, infant syrup containing ampicillin may be given in recommended doses.

Medical help should be sought if diarrhea lasts more than four days without improvement, or more than one day in a small child with severe diarrhea. Help is also needed if dehydration worsens, if the child vomits everything or continues vomiting for more than three hours after starting rehydration drink, if seizures occur, if swelling of feet and face appears, if the person was already weak or malnourished, or if there is significant blood in the stool.

Vomiting is common in children and adults and is often mild and temporary. It may be caused by stomach infections, food poisoning, intestinal blockage, appendicitis, high fever, malaria, hepatitis, tonsillitis, ear infections, meningitis, urinary infection, gallbladder pain, or migraine.

Danger signs with vomiting include worsening dehydration, severe vomiting lasting more than 24 hours, green or brown vomit suggesting obstruction, constant abdominal pain with inability to pass stool, and vomiting blood. These require urgent medical attention.

For simple vomiting, avoid solid food while vomiting is severe. Small sips of cola, ginger ale, herbal teas, or rehydration drink can help prevent dehydration. If vomiting continues, medicines such as promethazine or diphenhydramine may be used in recommended doses. Medication should not be repeated until dehydration improves and urination resumes. In severe cases where oral or rectal medicine is not possible, an injection may be required under proper guidance.

Simple headaches can improve with rest and aspirin. Warm compresses and gentle massage of the neck and shoulders may help. Headache is common with fever. Severe headache should be checked for signs of meningitis. Recurrent headaches may indicate chronic illness or poor nutrition and require medical evaluation.

Migraine is a severe throbbing headache often affecting one side of the head. It may begin with visual disturbances or numbness, followed by intense pain and sometimes vomiting. Migraines are painful but not dangerous. At the first sign, take aspirin with strong coffee or tea and rest in a dark, quiet place. For severe attacks, aspirin with codeine or ergotamine with caffeine may be used in recommended doses. Ergotamine combinations should not be used during pregnancy.

Colds and the flu are common viral infections that usually cause a runny nose, cough, sore throat, fever, body pain, and sometimes joint pain. Young children may also have mild diarrhea. These illnesses usually go away without medicine. Antibiotics such as penicillin or tetracycline should not be used because they do not work against viruses and may cause harm. People with a cold should drink plenty of water, get enough rest, and may take aspirin or acetaminophen to reduce fever and body aches. Expensive cold medicines are not more effective than simple pain relievers. A sore throat is often part of a cold and usually does not need special medicine, though gargling with warm water may help. If a sore throat begins suddenly with high fever, it may be strep throat and requires special treatment. If a cold lasts more than a week or includes high fever, chest pain, fast breathing, or thick mucus with pus, it may develop into bronchitis or pneumonia. Older adults and people with lung problems are at greater risk of complications.

Colds spread from person to person through virus droplets in the air, not from being cold or wet. Getting enough sleep and eating well, including fruits rich in vitamin C, may help prevent colds. To avoid spreading infection, a sick person should cover the mouth and nose while coughing or sneezing and avoid close contact with babies. Blowing the nose too hard may lead to earache or sinus problems, so it is better to gently wipe the nose.

A stuffy or runny nose may occur due to a cold or allergy. Excess mucus can cause ear infections in children and sinus problems in adults. Clearing the nose with gentle suction in children, sniffing salt water, or breathing steam may help loosen mucus. Decongestant nose drops such as phenylephrine may be used for short periods but should not be used more than three times a day or for more than three days.

Sinusitis is inflammation of the sinuses and often occurs after a cold or throat infection. It may cause facial pain above or below the eyes, thick or bad-smelling mucus, stuffy nose, fever, and sometimes tooth pain. Treatment includes drinking fluids, breathing steam, using salt water in the nose, applying warm compresses, and sometimes taking antibiotics if prescribed. If symptoms do not improve, medical help should be sought.

Hay fever is an allergic reaction that causes a runny nose and itchy eyes. It is usually triggered by substances in the air such as pollen, dust, mold, or animal hair. Allergies are not infections and cannot spread from one person to another. However, they may run in families. Allergic reactions can include skin rashes, breathing difficulty, diarrhea in rare cases, or even severe allergic shock. Avoiding triggers and using antihistamines such as chlorpheniramine may help.

Asthma is a condition that causes repeated attacks of difficult breathing. A person may produce a wheezing sound while breathing and may struggle to get enough air. In severe cases, lips and nails may turn blue. Asthma often begins in childhood and may worsen during certain seasons, at night, or after exposure to allergens, smoke, or pollution. Emotional stress may also trigger attacks. Treatment includes moving to fresh air, staying calm, drinking fluids, breathing steam, and using a rescue inhaler such as salbutamol. People with frequent attacks may need a controller inhaler such as beclomethasone. In emergencies without inhalers, epinephrine may be injected. If fever is present or symptoms last more than three days, antibiotics may be required. Avoiding triggers and keeping living areas clean can help prevent attacks.